Solutions and Colligative Properties

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Solutions and Colligative Properties

In day to day life we rarely come across pure substances, from the air we breathe to the food we eat almost everything is a mixture. Mostly these mixtures contain two or more pure substances. The utility and importance of these substances depend on their composition.

For example, the properties of German silver (mixture of copper, zinc and nickel) are quite different from that of brass (mixture of copper and zinc); to prevents tooth decay 1 part per million (ppm) of fluoride ions in water, while a high concentrations of fluoride ion can be poisonous (eg. sodium fluoride is used in rat poison).

In this portion, we will mostly consider liquid solutions and formation of liquid solution. This will be done by studying the properties like vapour pressure and colligative properties, of the solutions.

Definitions

Solution: A solution is a homogeneous mixture of two or more chemically non-reacting substances whose composition can be varied within certain limits.

Solvent: A solvent is that component of the solution which is present in the largest amount by mass. A solvent determines the physical state of solution in which it exists.

Solute: One or more components present in the solution other than the solvent are called solutes.

Definitions

A solution containing only one solute dissolved in given solvent is called **Binary Solution**

Types of Binary Solutions:

	Solvent	Solute	Examples
1.	Gas	Gas	Mixture of gases, eg. air.
2.	Gas	Liquid	Water vapour in air, mist. $CHCl_3(l) + N_2(g)$
3.	Gas	Solid	Smoke, camphor (s) + $N_2(g)$.
4.	Liquid	Gas	CO_2 gas dissolve in water (aerated drink), soda water.
5.	Liquid	Liquid	Mixture of miscible liquids e.g. alcohol in water.
6.	Liquid	Solid	Salt in water, sugar in water.
7.	Solid	Gas	Hydrogen over palladium.
8.	Solid	Liquid	Mercury in zinc, mercury in gold i.e. all amalgams.
9.	Solid	Solid	Alloys e.g. copper in gold. zinc in copper.

Concentration Terms:

1. **Terms based on percentage:**

(a) % weight by weight(%w/w):

$$\%w/w = \frac{\text{weight of solute (gm)}}{\text{weight of solution (gm)}} \times 100$$

E.g. 10% w/w urea solution means, 10 gm of urea is present in 100 gm of solution **OR** 10 gm of urea is present in 90 gm of water.

(b) % weight by volume(%w/v)

$$\%\ w/v = \frac{\text{gm of solutes}}{\text{vol. of solution in mL}} \times 100$$

10% (w/v) urea solution means 10 gm of urea is present in 100 mL of solution.

Do You Know?

For very dilute solution: vol. solution = vol. solvent.

(c) % Volume/Volume:

$$\%\ v/v = \frac{\text{Volume of solute}}{\text{vol. of solution in mL}} \times 100$$

e.g. 10% v/v alcohol ethanol aq solution means 10 ml of ethanol in 100 ml of solution
10 ml of C_2H_5OH in 90 ml of H_2O.

Do You Know?

%V/V is used when both solute & solvent are liquids. You may notice on a sanitizer pack %V/V content of alcohol in the pack.

Definitions

% weight by weight(%w/w):
weight of solute dissolved in 100 gm of a solution.

Points to remember

%w/w does not depend on temperature.

Definitions

% Volume/Volume:
Volume, in (mL) of solute dissolved per 100 mL of solution.

Points to remember

10% (w/v) solution doesn't mean 10 gm of solute present in 90 ml of water

Definitions

% weight by volume(%w/v)
Grams of solute dissolved in 100 mL of solution.

2. Strength of solution in gm/L

$$gm/L = \frac{\text{weight of solute (gm)}}{\text{Volume of solution (inL)}}$$

$$\frac{\text{weight of solute (gm)}}{\text{Volume of solution (inml)}} \times 1000$$

Definitions

Strength of solution in gm/L
weight of solute (in gms) per litre
(1000 mL) of solution.

Q1 **10% (w/v) sucrose solution, then specify its conc. in gm/L**

Sol. In 100 mL ..... 10 gm sucrose is available

1000 mL solution will contain $\frac{10}{100} \times 1000 = 100$ gm/L

Q2 **If we have 6% w/w urea soln. with density 1.060 gm/mL then calculate its strength in g/L.**

Sol. 6 gm in 100 gm solution.

$\frac{100}{1.060}$ mL $\longrightarrow$ 6 gm.

1000 mL = $\frac{6}{100} \times 1.060 \times 1000 = 10.6 \times 6 = 63.6$ gm/L

3. Molarity

Assume that
Moles of Solute = n
Moles of Solvent = N

$$M = \frac{n}{V(\text{in L})} = \left(\frac{W}{M}\right) \times \frac{1000}{V \text{ in (mL)}}$$

no. of moles of solute = molarity x volume (in L)
no. of mili moles of solute = molarity x volume (in mL)

Definitions

Molarity
Number of moles of solute dissolved per litre of solution.

Points to remember

Molarity depends on temperature as volume depends on temperature

Do You Know?

V_1 ml of C_1M solution is mixed with V_2 mL of C_2M solution (same substance or solute)

C_f= final concentration

$$C_f (V_1 + V_2) = C_1V_1 + C_2V_2$$

$$C_f = \left[\frac{C_1V_1 + C_2V_2}{V_1 + V_2}\right] = \frac{\text{Total moles}}{\text{Total volume}}$$

Q3 **If 20 ml of 0.5 M Na_2SO_4 is mixed with 50 ml of 0.2 M H_2SO_4 & 30 ml of 0.4 M $Al_2(SO_4)_3$ solution calculate. $[Na^+]$, $[H^+]$, $[SO_4^{2-}]$, $[Al^{3+}]$. Assuming 100% dissociation**

Sol. $M = \frac{\text{moles}}{\text{volume}} = 10$ m. moles of Na_2SO_4

$\Rightarrow$ 20 m. moles of Na^+

(i) $[Na^+] = \frac{20}{100} = 0.2$ M

(ii) $[H^+]$ = ?

10 m. moles H_2SO_4

20 m. moles H^+

$[H^+] = \frac{20}{100} = 0.2$ M

(iii) $SO_4^{2-} = \frac{10 + 10 + 36}{100} = \frac{56}{100} = 0.56$M

(iv) $Al^{3+} = \frac{24}{100} = 0.24$ M

4. Molality

If w gm of solute (Molar mass = m gm/mole) is dissolved in 'W' of solvent.

$$\text{molality} = \left(\frac{w}{M}\right) \times \frac{1000}{W(gm)}$$

Where M is Molar mass of Solute.

$$\text{molality} = \frac{\text{moles} \times 1000}{\text{weight of solvent(gm)}}$$

Definitions

Molality

Number of moles of solute dissolved per 1 kg (1000 gm) of solvent.

Points to remember

Relationship between molality & molarity

Assume that ***w*** gm of solute of molar mass **m** gm/mol is dissolved in **W** gm solvent & density of resulting solution = **'d'** gm/ml.

say 1 L solution taken,

mass of 1 lit solution = (1000 x d) gm

moles of solute = (molarity)

mass of solute = (molarity) x m

mass of solvent = W = 1000 d — (molarity) x m

$$\text{molality} = \frac{(\text{molarity}) \times 1000}{1000d - \text{molarity} \times \text{M.Wt}}$$

Q4 **Calculate molarity of 1.2 M H_2SO_4 solution if its density = 1.4 gm/mL**

Sol. $\text{Molarity} = \frac{1.2 \times 1000}{1000 \times 1.4 - 1.2 \times 98} = 0.935\text{M}$

5. Normality

$$N = \frac{\text{no. of equivalents of solute}}{\text{volume of solution(in L)}}$$

no. of equivalents = normality × volume (in L)

(Normality = n × molarity)

Where n= 'n' factor

Definitions

No. of equivalents of solute dissolved per litre of solution

How to calculate n-factor:

(i) For oxidizing/reducing agents :

no. of e^- involved in oxidation / reducion half reaction per mole of O.A./R.A.

$5e^- + 8H^+ + MnO_4^- \longrightarrow Mn^{2+} + H_2O$

(ii) For acid/ base reactions:

no. of H^+ ions displaced/ OH^- ions displaced per mole of acid/ base.

(iii) For salt:

$$\left.\begin{array}{c} n = \text{Total charge on cations.} \\ \text{or} \\ \text{Total charge on anions} \end{array}\right\} \text{Simple salts}$$

6. Mole Fraction

$$X_{solute} = \frac{\text{moles of solute}}{\text{total moles in solutions}} = \frac{n}{n+N}$$

$$X_{Solvent} = \frac{\text{Moles of solvent}}{n+N}$$

$X_{solute} + X_{solvent} = 1$ (for binary solution)

Previous Year's Questions

The mole fraction of a solvent in aqueous solution of a solute is 0.8. The molality (in $molkg^{-1}$) of the aqueous solution is

(A) 13.88×10^{-2} (B) 13.88×10^{-1}

(C) 13.88 (D) 13.88×10^{-3}

Ans.: (C) is correct

Do You Know?

Relationship between Molality and Moles

$$\text{Molality}(m) = \frac{\text{Mass of solute } (w_2) \times 1000}{\text{Molar mass of solute } (M_2) \times \text{mass of solvent}(w_1)}$$

$$m = \frac{w_1}{M_1} \times \frac{1000}{w_1} \qquad m = n_2 \frac{1000}{n_1 \times M_1}$$

Q5 **If we have 10 molal urea solution calculate**

(i) Mole faction of urea in this solution & also calculate % w/w of urea. (MW = 60)

(ii) Molarity of solution (d=1.5 g/ml)

Sol. 10 moles urea in 1000 gm of water

$$X_{urea} = \frac{10}{10 + \frac{1000}{18}} = \frac{10}{65.55} \text{ (mole fraction)}$$

% w/w

weight of urea = 10 × 60 = 600

$$= \frac{10 \times 60}{10 \times 60 + 1000} \times 100\% = 37.5\%$$

(ii) d = 1.5 gm/mL . . molarity = % w/v & strength in gm/L of the solution

10 moles urea in 1000gm H_2O = (1000 + 600) gm of solution

$$= \frac{1600}{1.5} \text{ mL of solution}$$

$$\text{molarity} = \frac{10 \times 1.5}{1600} \times 1000 = \frac{150}{16} = 9.3M$$

Strength in gm/L = (9.3 x 60) gm/L

$$\% \text{ w/v} = \frac{600 \times 1.5}{1600} \times 100\% = \frac{60 \times 15}{16} = \frac{225}{4} = 56.25\%$$

Points to remember

For dil. aq. solution, molality ≈ molarity

$$\text{molarity} = \frac{\text{molarity} \times 1000}{1000 \times d - \text{molarity} \times m} \approx \text{molarity}$$

7. ppm (Parts Per Million).

(a) ppm(w/w)

$$= \frac{\text{wt. of solute (in gm)}}{\text{wt. of soltuion (in gm)}} \times 10^6 = 1 \text{ million}$$

(b) ppm (w/v) = $\frac{\text{wt. of solute (in gm)}}{\text{vol. of soltuion (in mL)}} \times 10^6$

(c) ppm (moles/moles) $\frac{\text{moles of solute}}{\text{moles of solution}} \times 10^6$

Solubility

Solubility of a solid in a liquid:

A given solution can not dissolve every solid. While naphthalene and anthracene do not dissolve in water sodium chloride and sugar dissolve readily. On the other hand, naphthalene & anthracene dissolve readily in benzene but sodium chloride and sugar do not..

Do You Know?

It is observed that polar solutes dissolve in polar solvents and non-polar solutes in non-polar solvents. In general, a solute dissolve in a solvent if the intermolecular interactions are similar in the two or we may say **like dissolves like.**

Definitions

Solubility of a substance is its maximum amount that can be dissolved in a specified amount of solvent at a specified temperature. It depends upon the nature of solute and solvent as well as temperature and pressure. Let us consider the effect of these factors in solution of a solid or a gas in a liquid.

- On adding a solid solute into the solvent, some of solute particles dissolve and the concentration increases in solution. This is called **dissolution**.
- The collision of solute particles with one another result in their separation from solution. This is called **crystallisation.**
- A dynamic equilibrium is attained between dissolution and crystallization, at this state change in concentration in solute is not seen under given temperature and pressure.



Solute + Solvent ⇌ Solution

At this stage the concentration of solute in solution will remain constant under the given conditions, i.e., temperature and pressure. Similar process is followed when gases are dissolved in liquid solvents.

Points to remember

- A solution in which we can not dissolve additional amount of solutes is called saturated solution.
- A solution in which additional amount of solute can be dissolved at given temperature is called unsaturated solution.

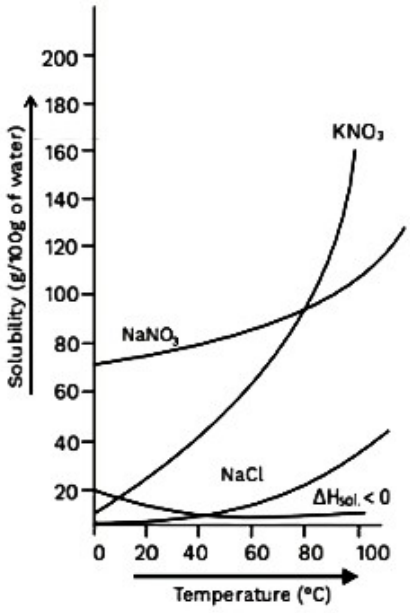
The solution which is in dynamic equilibrium with undissolved solute is the saturated solution and contains the maximum amount of solute dissolved in a given amount of solvent. Thus, the concentration of solute in such a solution is its solubility. Earlier we have observed that solubility of one substance into another depends on the nature of the substances. In addition to these variables, two other parameters, i.e., temperature and pressure also control this phenomenon.

Factors affecting solubility of solid in liquid solution:

Effect of Temperature:

The solubility of solids is largely affected by change in temperature. The solubility, a dynamic equilibrium, follows Le Chateliers Principle. If in a solution, the dissolution process is endothermic (Δ_{Sol} H > 0), the solubility should increase with rise in temperature and if it is exothermic (Δ_{Sol} H > 0) the solubility should decrease. These trends are also observed experimentally.

Solubility of Solids in Solution vs T



Effect of Pressure:

Solids and liquids are highly incompressible and their solubility is not much affected by change in pressure.

Solubility of a gas in a liquid (Henry's Law)

Henry law deal with effect of pressure on the solubility of gas.

Let X = Mole fraction of gas at a given temperature as a measure of its solubility.

p = Partial pressure gas in equilibrium with the solution.

According to Henry's law: $X \propto p$
or $p \propto X$
or $p = K_H X$

where K_H = Henry law constant.

Statement:
The solubility of a gas in a liquid at a given temperature is directly proportional to the pressure at which it is dissolved.

Characteristics of Henry law constant (K_H).

(i) unit same as those of pressure torr or Kbar.
(ii) Different gas have different value of K_H.
(iii) The K_H value of a gas is different in different solvents and it increase with the increase in temperature.
(iv) Higher the value of K_H of a gas lower will be its solubility.

Plot of p Vs X is a straight line passing through the origin with slop equal to K_H

Value of Henry's Law Constant for Some Selected Gases in Water

Gas	Temperature/K	K_H/kbar
He	293	144.97
H_2	293	69.16
N_2	293	76.48
N_2	303	88.84
O_2	293	34.86
O_2	303	46.82
Argon	298	4.03
CO_2	298	1.67
Formaldehyde	298	183×10^{-5}
Methane	298	0.413
Vinyl chloride	298	0.611

Points to remember

(i) If a mixture of gases is brought in contact with solvent each constituent gas dissolves in proportion to its partial pressure. It means Henry's law applies to each gas independent of the pressure of other gases.

(ii) Henry's law can also be applied while expressing the solubility of the gas in terms of mass per unit volume. Mass of the gas dissolved per unit volume of a solvent at a given temperature is directly proportional to the pressure of a gas in equilibrium with the solution.

$$m \propto p,$$

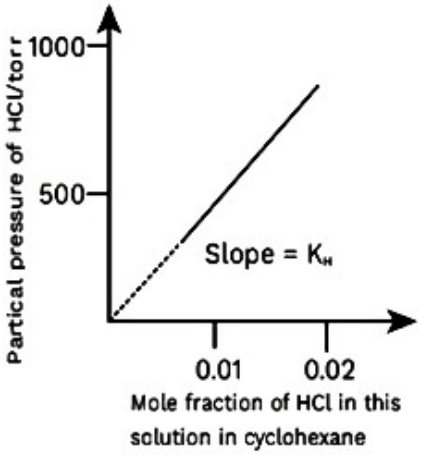
$$m = K \times p$$

where,

m = mass of gas dissolved in unit volume of solvent.

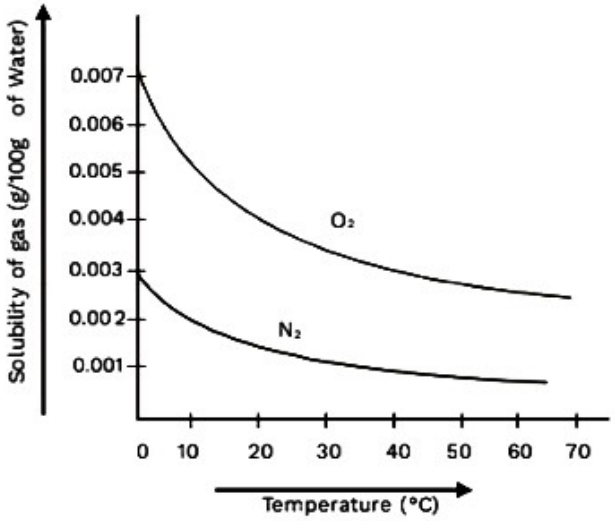
p = pressure of gas in equilibrium with solution. Where K is the constant of proportional that depends on nature of gas, temperature & unit of pressure.

Plot of p Vs X for solution of HCl in cyclohexane.



Effect of temperature on solubility of gas in liquid:

With rise in temperature solubility of gases in liquids decreases. While dissolved, the gas molecules are present in liquid phase and the process of dissolution can be considered similar to condensation and heat is evolved in this process. We have learnt that the dissolution process is dynamic equilibrium and thus follows Le Chatelier's principle. Since dissolution is an exothermic process, the solubility should decrease with increase of temperature.



Points to remember

K_H values for both N_2 and O_2 increase with increase of temperature indicating that the solubility of gases increases with decrease of temperature. It is due to this reason that aquatic species are more comfortable in cold water rather than warm water.

Application of Henry's law:

Several applications in biological and industrial phenomena.

(i) To increase the solubility of CO_2 in soft drinks and soda water the bottle is sealed under high pressure.

(ii) Pressure increases the solubility of atmospheric gases in blood, scuba divers must cope with high concentrations of dissolved gases while breathing air at high pressure underwater. When the divers come out on the surface, the pressure gradually decreases. Due to this sudden release of the dissolved gases take place and the formation of bubbles of nitrogen in the blood as well. This creates a painful medical condition known as bends. The diver tanks are filled with air diluted with helium (11.7% helium, 56.2% nitrogen and 32.1% oxygen) to avoid bends, as well as, the toxic effects of high concentrations of nitrogen in the blood.

Previous Year's Questions

The Henry's law constant for the solubility of N_2 gas in water at 298 K is 1.0×10^5 atm. The mole fraction of N_2 in air is 0.8. The number of moles of N_2 from air dissolved in 10 moles of water of 298 K and 5 atm pressure is:

(A) 4×10^{-4} (B) 4.0×10^{-5}

(C) 5.0×10^{-4} (D) 4.0×10^{-6}

Ans. (A) is correct

(iii) At higher altitudes the partial pressure of oxygen is less as compared to the ground level. The amount of oxygen in the blood and tissues of people living at high altitudes or climbers decreases due to this. Fall in blood oxygen level causes climbers to become weak and unable to think clearly, symptoms of a condition known as anoxia.

Question:

Which one of the following statements regarding Henry's law is not correct?

(a) Different gases have different K_H (Henry's law constant) values at the same temperature

(b) Higher the value of K_H at a given pressure, higher is the solubility of the gas in the liquids

(c) The value of K_H increases with increase of temperature and is function of the nature of the gas.

(d) The partial pressure of the gas in vapour phase is proportional to the mole fraction of the gas in the solution

Ans.: (B) is correct

Limitation of Henry's law:

Henry's law is valid only in following condition.

(i) The pressure of gas should not be too high.

(ii) The temperature is not too low.

(iii) The gas should not go any chemical reaction with the solvent.

(iv) The gas should not undergo dissociation in solution.

Q6 **Over a solution containing 6.56×10^{-3} g of ethane has 1 bar partial pressure of ethane. Find the partial pressure of the gas, given that the solution contains 5.00×10^{-2} g of ethane, then?**

Sol. 7.62 bar

$$\frac{6.56 \times 10^{-3}}{5 \times 10^{-2}} = \frac{1}{P} \qquad \Rightarrow P = 7.62 \text{ bar.}$$

Q7 **At 30°C Benzene and toluene have vapour pressure in the pure state 119.6 mm and 36.7 mm respectively. What are the partial vapour pressure of each of these components and the total pressure over a solution consisting of equal weights of these two components at 30°C? What is the composition of the vapour in equilibrium with the solution at this temperature? Comments the results.**

Sol. Partial pressure of benzene = $P_b^o \times X_b$

$\because \boxed{w_b = w_t}$

$$= \frac{119.6 \times \frac{w_b}{78}}{\frac{w_b}{78} + \frac{w_t}{92}}$$

$= 64.85$ mm

$= 64.85$ mm

Partial Pressure of toluene $= P_t^0 \times X_t$

$$= 36.7 \times \frac{\frac{w_t}{92}}{\frac{w_b}{78} + \frac{w_t}{92}}$$

$=16.90$

Total pressure = 16.90 + 64.85 = 81.75 mm

Mole fraction in solution of benzene

$$= \frac{64.85}{81.75} = 0.206$$

Mole fraction in solution of toluene

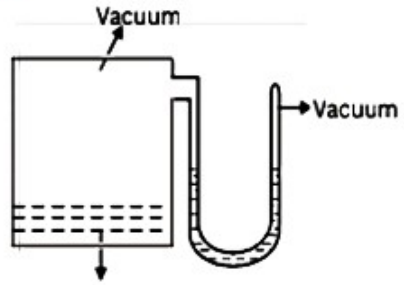
$$= \frac{16.90}{81.75} = 0.793$$

Q8 **An organic liquid, A, is immiscible with water. When boiled together with water, the boiling point is 90°C at which the partial vapour pressure of water is 526 mm Hg. The atmospheric pressure is 736 mm Hg. The weight ratio of the liquid and water collected is 2.5: 1. What is the molecular weight of the liquid ?**

Sol. $\frac{P^o - P_s}{P_s} = \frac{n}{N}, \frac{736 - 526}{526} = \frac{2.5}{m} \times \frac{18}{2}$

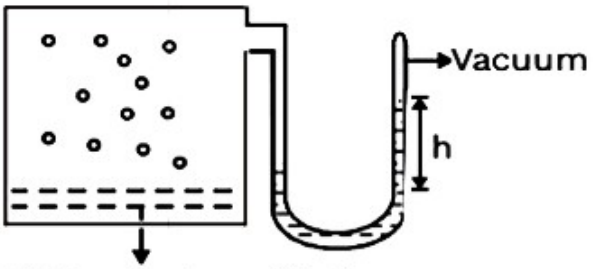
$\boxed{m = 112.7 \text{ gm}}$

Vapour Pressure



liquid (10gm) → t = 0

(Imagine no liquid molecule exist in vapour phase)



liquid (5 gm) at equilibrium

say 5 gm liquid left at t_{eq}

At equilibrium : the rate of evaporation = rate of condense

$H_2O(l) \rightleftharpoons H_2O(g)$

$K_p = P_{H_2O(g)}$

Hence V.P is an equilibrium constant (K_p) of the reaction.

liquid ⇌ vapours.

Definitions

The pressure exerted by vapours of the liquid when equilibrium is established between vapours & the liquid.

Points to remember

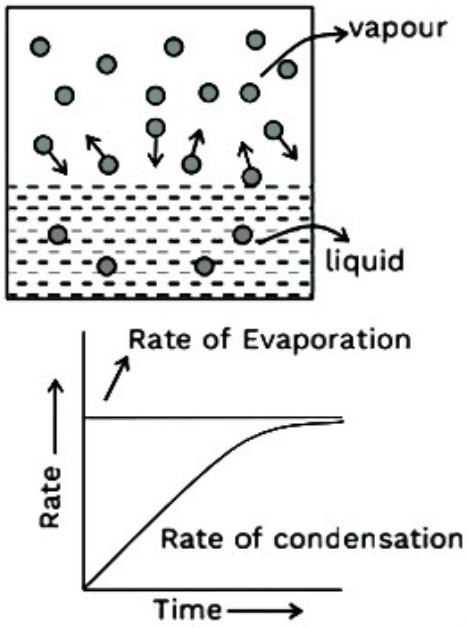
Since vapour pressure is an equilibrium constant so its value is dependent only on temp. for a particular liquid not only on the amount of liquid taken on surface area of the liquid or on volume or shape of the container.

Let's understand

(i) evaporation
(ii) condensation

During **evaporation** from the surface particles continue to break away from the surface of the liquid. As the gaseous particles bounce around, some of them will hit the surface of the liquid again, and will be trapped there. **This is called condensation**. The rate of condensation increases with time, but rate of evaporation remains constant. There will be an equilibrium set up, rapidly, in which the number of particles leaving the surface is exactly balanced by the number re-joining it.

This pressure at equilibrium is called the **saturated vapour pressure** (also known as **saturation vapour pressure**).



Factors Influencing Vapour pressure

(1) Temperature

(2) Nature of liquid

1. Effect of Temperature on vapour pressure

While the saturation between the space above the liquid with vapour particles is attained, you have this equilibrium occurring on the surface of the liquid:

$$\text{liquid} \rightleftharpoons \text{vapour}, \Delta H > 0$$

Points to remember

The dependence of vapour pressure with temperature is given by **Clausius-Clapeyron equation**

$$\ln\left(\frac{p_2}{p_1}\right) = -\frac{\Delta H_v}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right),$$

where ΔH_V = molar enthalpy of vapourisation of the given liquid, p_2 = vapour pressure of the liquid at T_2 and p_1 = vapour pressure of the liquid at T_1.

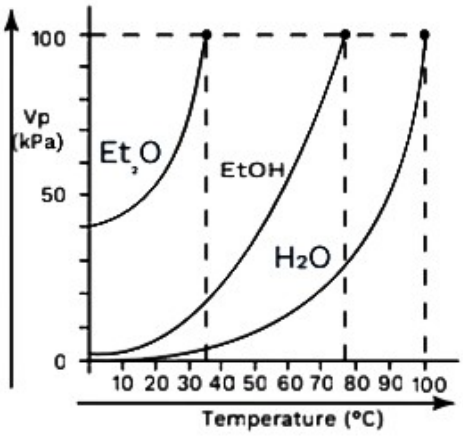
Points to remember

- The partial pressure of vapours in equilibrium with pure solid or pure liquid at a given temperature. At eq. the rate of evaporation = rate of condensation.

$$\text{e.g. } H_2O(g) \rightleftharpoons H_2O(g)$$

$$K_p = P^{eq}_{H_2O(g)}$$

- Vapour pressure of water is called aqueous tension.



Points to remember

Vapour pressure of a liquid does not depend on

(i) the amount of liquid taken

(ii) surface area of the liquid

(iii) volume or shape of the container

The forward change (liquid to vapour) is endothermic. It needs heat to convert the liquid into the vapour. According to Le Chatelier, increasing the temperature of a system in a dynamic equilibrium favours the endothermic change. That means increasing the temperature increase the amount of vapour present, and so increases the saturated vapour pressure.

Effect of nature :

With increase in the force of attraction between the particles of liquid, the vapour pressure decreases.

$$V.P. \propto \frac{1}{\text{Inter molecular attraction force (I.M.A.F.)}}$$

e.g. $n\text{-}C_6H_{14} > n\text{-}C_7H_{16} > C_8H_{18}$

Important Points related to vapour pressure:

The partial pressure of vapours in equilibrium with pure solid or pure liquid at a given temperature. At equilibrium, the rate of evaporation = rate of condensation

Example: $H_2O(g) \rightleftharpoons H_2O(g)$

$K_p = p^{eq}_{H_2O,g}$

i) Vapour pressure of water is called aq. Tension

ii) Relative Humidity (R.H.) $= \frac{\text{Partial pressure of water vap.}}{\text{v.p. of water at same temp.}} \times 100$

(i) Boiling point: The temperature at which the bulk of liquid, may convert to vapor, on suppying heat at a given pressure.

ii) At boiling pt. $P^\circ = P_{ext}$

iii) At normal boiling pt $P^\circ = 1$ atm

i) Since vapour pressure is an equilibrium constant so it's value is dependent only on
   a) temperature for a particular liquid.
   b) Nature of liquid.

ii) It does not depend on
   a) Amount of liquid
   b) Surface area of liquid
   c) Volume or shape of container

Moreover, we can say that it is a characteristic constant.

The numerical value of liquid's vapour Pressure depends on the magnitude of the intermolecular force present and on temperature.

i) The smaller the intermolecular force, higher the vapour pressure because loosely held molecules escape more easily.

ii) The higher the temperature, higher the vap. Pressure because larger fraction of molecules have sufficient kinetic energy to escape. (Acc. To maxwell distribution law

Vapour pressure represents the highest partial pressure which may be exerted by vapours of liquid at that temperature.

If it's partial pressure is increased beyond this value (P°) its vapours begin to condense till their partial pressure becomes equal to the vapour pressure.

If the partial pressure of vapour is less than v.p. of liquid the liquid (if present) will vaporize till.

(a) its v.p. is attained or

(b) the liquid completely gets vaporized

If a volatile solid/liquid is brought in constant with a gas (or vacuum), vapours of that solid/liquid occupy the gas phase till the gas phase becomes saturated with that solid/ liquid vapours.

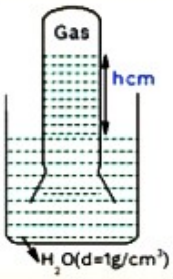
Q9 **The v.p. of water at 80°C is 355 mm of Hg(torr). A 100 ml vessel saturated solution of oxygen in water at 80°C, the total gas pressure being 760 mm Hg(torr). The contents of the vessel were shifted into a 50.0 ml, vessel keeping temperature 80° C. What were the partial pressures of oxygen and of water vapor, the total pressure in the final state of equilibrium ? Neglect changes in the volume due to any water which might condense.**

Sol. P_{O_2} = 810mm Hg, P_{H_2O} = 355 mm Hg , P_{total} = 1165 mm Hg

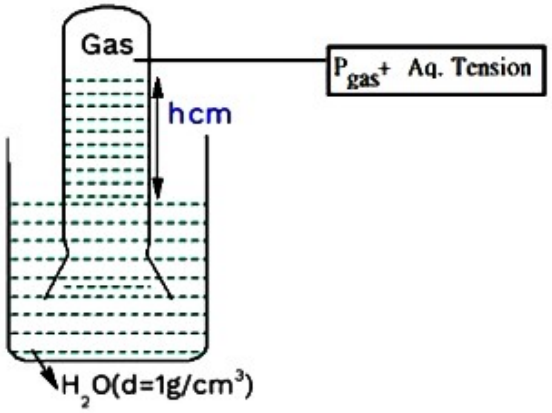
In 100 ml vessel which contained water - saturated oxygen, the pressure of O_2 gas = 760 – 355 = 405 torr

when the contents of this vessel were pumped into 50 ml vessel, at the same temperature, the pressure of oxygen gets doubled i.e. P_{O_2} = 810 torr. But pressure of water vapour will remain constant, as some vapour in this 50 ml vessel, gets condensed. So P_{H_2O} = 355 torr & Total pressure = 810 + 355 = 1165 torr.

Q10 **An ideal gas is accumulated by downward displace ment of water. Select the correct expression for P_{gas} (in Torr) according to the diagram [dHg = 13.6 g /cm³]:**



Sol. The atmospheric pressure is balanced by the sum of the pressures of gas, aqueous tension(Vapour Pressure) of water and water column of height **h**.



Thus, $P_{atmp} = P_{gas} + \left[\text{aq. Tension} + \frac{h}{13.6}\right]$

$P_{gas} = P_{atmp} - \left[\text{aq. Tension} + \frac{h}{13.6}\right]$

Raoult's Law

Chemist François-Marie Raoult discovered that when a non volatile substance is dissolved in a solution, the vapor pressure of the solution generally decreases. When we apply this to a liquid - liquid solution it may decrease or increase depending on vapour pressure of the substances being mixed.

Points to remember

In our scope of scope of study raoult's law is used only for a binary solution

Solutions Containing

$\left\{\begin{matrix}\text{VOLATILE SOLVENT}\\ \text{VOLATILE SOLUTE}\end{matrix}\right\}$ Liq. solution

Raoult's law (for volatile liquid Mixture)

let A, B two volatile liquids mixed together

Partial pressure of A = P_A & According to Raoult's law (experimentally)

$P_A \propto X_A$

$P_A = X_A P_A^o$

P_A^o = V.P. of pure liq. A

(constant at a particular temp)

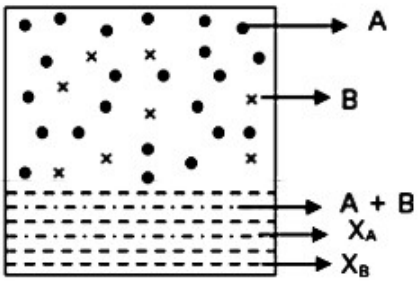
Similarly, $P_B \propto X_B$

$\therefore P_B = X_B P_B^o$ (P_B^o = V.P of pure liquid B.)

$\therefore$ According to Dalton's law

Final equation for vapour pressure of solution will be

$$\boxed{P_T = P_A + P_B = X_A P_A^o + X_B P_B^o}$$



{mole factions of A & B in liq. solution}

Points to remember

if $P_A^o > P_B^o$

$\therefore$ A is more volatile than B

$\therefore$ B.P. of A < B.P. of B

Calculation of amount of liquids in vapour phase

X'_A = mole fraction of A in vapour above the liquid/solution.

X'_B = mole fraction of B in vapour above the liquid/solution.

Dalton's law for gaseous mix

Pressure of A in gas phase = $X'_A P_T$

Raoult's law when liquid & vapour In equilibrium
Both liquid will be in equilibrium with their respectively gas phase

$$P_A = X_A P_A^\circ \qquad X'_A P_T = X_A P_A^\circ$$

$$P_B = X'_B P_T = X_B P_B^\circ$$

$$X_A + X_B = 1 = \frac{X'_A P_T}{P_A^\circ} + \frac{X'_B P_T}{P_B^\circ}$$

$$\boxed{\frac{1}{P_T} = \frac{X'_A}{P_A^\circ} + \frac{X'_B}{P_B^\circ}}$$

Graphical representation of Raoult's law:

$$P_A = X_A P_A^\circ \;\&\; P_B = X_B P_B^\circ$$

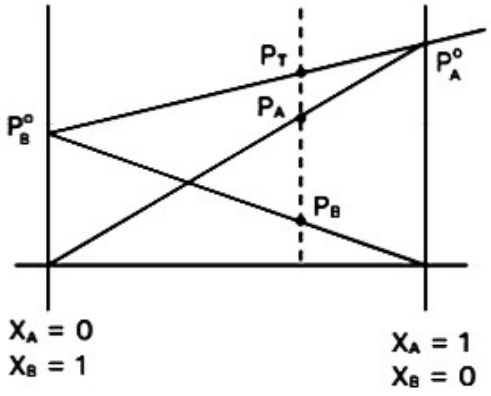
$$P_T = X_A P_A^\circ + X_B P_B^\circ$$

$$P_T = X_A P_A^\circ + (1 - x_A) P_B^\circ = \left(P_A^\circ - P_B^\circ\right) X_A + P_B^\circ$$

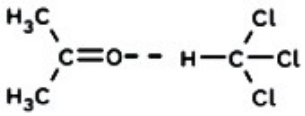
$$P_T = (1 - x_B) P_A^\circ + X_B P_B^\circ = \left(P_B^\circ - P_A^\circ\right) X_B + P_A^\circ$$

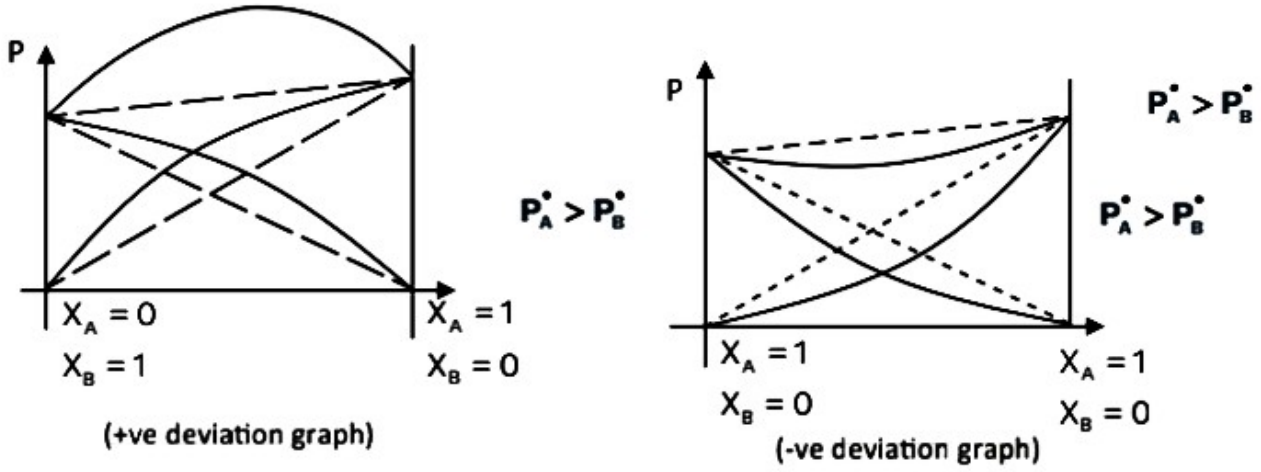
A more volatile than

$$P_T = P_A + P_B.$$

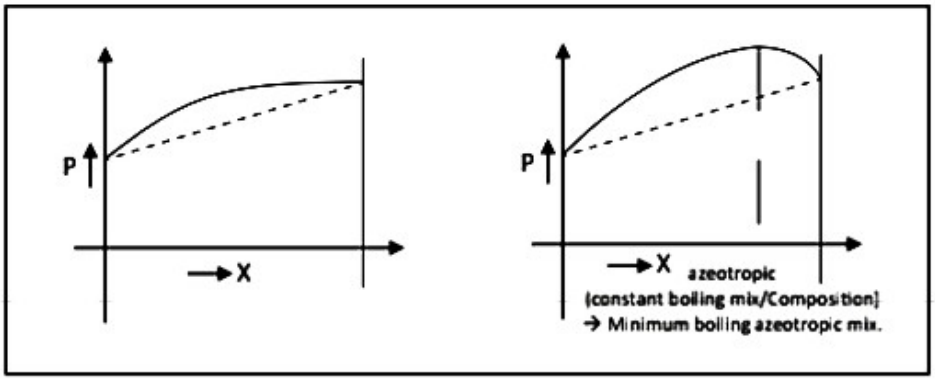


Ideal and Non-ideal Solution

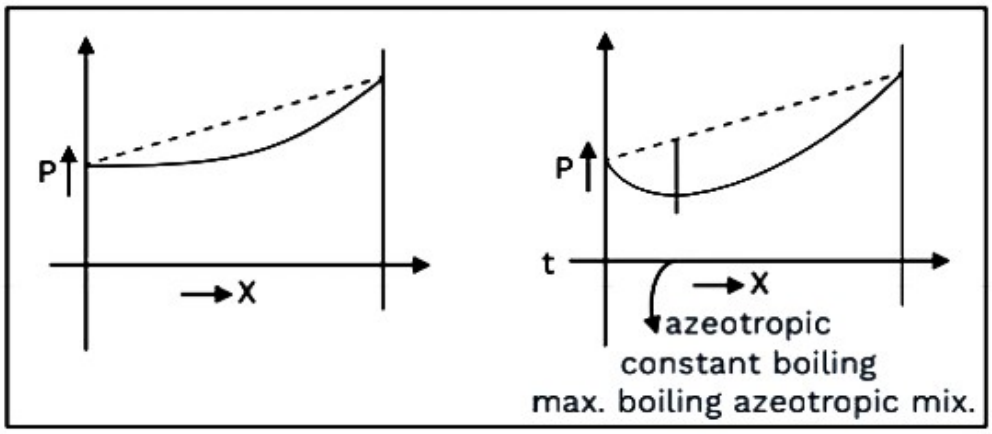
Ideal solutions (mixtures):	Non-Ideal solutions:	
	Positive deviation	**Negative deviation**
i) Which follow Raoult's law at all temp. & at all compositions.	Which do not follow Rault's law	Which do not obey Raoult's law.
$P_T = X_A P_A^\circ + X_B P_B^\circ$	(i) $P_{T,exp} > \left(X_A P_A^\circ + X_B P_B^\circ\right)$	(i) P_T exp $< x_A p_A^\circ + x_B p_B^\circ$
ii) Force of attraction b/w the liq. Molecules are exactly same.	(ii) $\begin{matrix} A---A \\ B---B \end{matrix} > A---B$ ↓ Weaker force of attraction	(ii) $\begin{matrix} A---A \\ B---B \end{matrix} > A---B.$ strong force of altraction.
A---A = A -----B = B---B	(iii) ΔH_{mix} = +ve energy absorbed	(iii) ΔH_{mix} = −ve (release energy)
iii) $\Delta H_{mix} = 0$	(iv) ΔV_{mix} = +ve (1L+1L>2L)	(iv) ΔV_{mix} = −ve (1L + 1L < 2L)
iv) $\Delta V_{mix} = 0$	(v) ΔS_{mix} = +ve	(v) ΔS_{mix} = +ve
v) ΔS_{mix} = +ve as for process to proceed.	(vi) ΔG_{mix} = −ve.	(vi) ΔG_{mix} = −ve.
vi) ΔG_{mix} = −ve	e.g. (1) $H_2O + CH_3OH$.	e.g. (1) $H_2O + HCOOH$
e.g. (1) Benzene + Toluene.	(2) $H_2O + C_2H_5OH$	(2) $H_2O + CH_3COOH$
(2) Hexane + heptane.	(3) C_2H_5OH + hexane	(3) $H_2O + KNO_3$
(3) $C_2H_5Br + C_2H_5I$.	(4) C_2H_5OH + cyclohexane.	(4) $CHCl_3 + CH_3OCH_3$
	(5) $CHCl_3 + CCl_4$ → dipole dipole interaction becomes weak	



Positive Deviation



Negative Deviation



Q11 **An equimolar mix of benzene and toluene is prepared the total V.P. of this mix as a function of mole fraction of benzene is found to benzene is found to be $P_T = 200 + 400X_{ben}$.**

(a) Calculate composition of vapours of this mix. [Assume that the no. of moles going into vapour phase is negligible in comparison to no. of moles present in liq. phase].

(b) If the vapour above liq. in part A are collected and are condensed into a new liquid calculate composition of vapours of this new liq.

Sol. (a) $P^{\circ}_{Benzene} = 600$ mm of Hg

$P^{\circ}_{Toluene} = 200$ mm of Hg

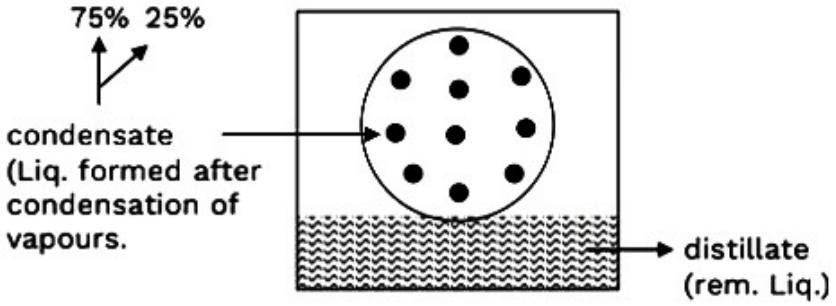
$$P_T = \frac{1}{2} \times 600 + \frac{1}{2} \times 200 = 400 \text{ mm of Hg}$$

$P_{benz} = X_{ben} \cdot P^{\circ}_{ben} = X'_{ben} P_T$.

$$X'_{benzene} = \frac{1/2 \times 600}{400} = \frac{3}{4} = 75\%$$

$$X'_{benzene} = \frac{1}{4} = 25\%$$

(b)



$$P_T = \frac{3}{4} \times 600 + \frac{1}{4} \times 200 = 500 \Rightarrow x_{Ben} = \frac{3/4 \times 600}{500} = 0.9 = 90\%$$

$X'_{Tol} = 0.1 = 10\%$

This process is known as fractional distillation.

Q12 **1 mol of benzene and 1 mol of toluene are mixed together to form a liquid mixture.**

(a) At what pressure does the first bubble is formed when pressure is re duced?

(b) Composition of the first bubble formed will be?

(c) If the pressure is reduced, at what pressure does the last trace of liquid disappear?

(d) Find the composition of the last drop of liquid?

(e) What will be the pressure, composition of the liquid and the composition of vapour, when 1 mol of the mixture has been vaporized? Given P_T° = 40 mm Hg, P_B° = 100 mm Hg

Sol. (a) 70 mm Hg, (b) $x_b = 5/7$, (c) P = 400/7 = 57.14 mm

(a) $P = X_A P_A^c + X_B P_B^c = 0.5 \times 40 + 0.5 \times 100 = 70$

(b) $Y_A = \dfrac{0.5 \times 40}{70} = \dfrac{2}{7}$ $\quad Y_B = \dfrac{5}{7}$

(c) at last trace of liquid

$Y_A = 0.5 \quad Y_B = 0.5$

$$\frac{1}{P} = \frac{Y_A}{P_A^\circ} + \frac{Y_B}{P_B^\circ} = \frac{0.5}{40} + \frac{0.5}{100}$$

$$P = \frac{400}{7}$$

(d) $Y_A = \dfrac{X_A P_A^\circ}{P} \Rightarrow 0.5 = \dfrac{X_A 40}{400/7} = \dfrac{5}{7}$; $X_B = \dfrac{2}{7}$

(e)

	Mole of 'a'	Mole of 'b'
Liquid	(1-x)	x
Vapour	x	(1-x)

P= 40(1-x) + 100x

$$\frac{1}{P} = \frac{Y_A}{P_A^\circ} + \frac{Y_A}{P_B} \Rightarrow \frac{1}{P} = \frac{x}{40} + \frac{1-x}{100} = \frac{100x + (1-x)40}{40 \times 100}$$

So, $p^2 = 40 \times 100$

$p = 20\sqrt{10}$

$20\sqrt{10} = 40(1-x) + 100x = 40 + 60x$

$$x = \frac{20\sqrt{10} - 40}{60} = \frac{\sqrt{10} - 2}{3}$$

Q13 **The vapour pressure of two miscible liquids A and B are 300 and 500 mm of Hg respectively. In a flask 10 moles of A is mixed with 12 moles of B. However as soon as B is added, A starts polymerising into a completely insoluble solid. The polymerisation follows first order kinetics. After 100 minutes, 0.525 mole of a solute is dissolved which arrests the polymerisation completely. The final vapour pressure of the solution is 400 mm of Hg. Estimate the rate constant of the polymerisation reaction. Assume negligible value change on mixing and polymerisation, and ideal behaviour for the final solution.**

Sol.

$P_A^\circ = 300$ mm $\qquad P_B^\circ = 500$ mm

nA = 10 mol $\qquad$ nB = 12 mol

Let the amount of A after 100 min is reduced to n_A. At this stage.

$$n_{Total} = n_A + n_B + n_{solute}$$
$$= n_A + 12 + 0.52$$
$$= n_A + 12.52 \text{ mol}$$

the amount fractions of A and B in the solution will be.

$$x_A = \frac{n_A}{n_A + 12.52\text{mol}} \text{ and } \quad x_B = \frac{n_A}{n_A + 12.52\text{mol}}$$

$$Pt = P_A^\circ x_A + P_B^\circ x_B$$

$$400 = \left(\frac{n_A}{n_A + 12.52}\right)300 + \left(\frac{12}{n_A + 12.52}\right)$$

$n_A = 9.90$ ml

For first order kinetics,

$$\ln \frac{[A]_t}{[A]_o} = -kt$$

$$K = -\frac{\ell n [A]_t / [A]_o}{t} = \frac{\ell n (9.90/10)}{100\text{min}}$$

$= 1.005 \times 10^{-4}$ min^{-1}.

Applications of Raoult's Law:

1. **Phase Diagrams of Two-Component Ideal Solutions: Bubble And Dew Points**

The compositions of the liquid and vapour that are in mutual equilibrium are not necessarily the same. Common sense suggests that the vapour should be richer in the more volatile component. This can be easily derived from Raoults's Law.

$$\frac{1}{P_T} = \frac{y_A}{P_A^0} + \frac{y_B}{P_B^0}$$

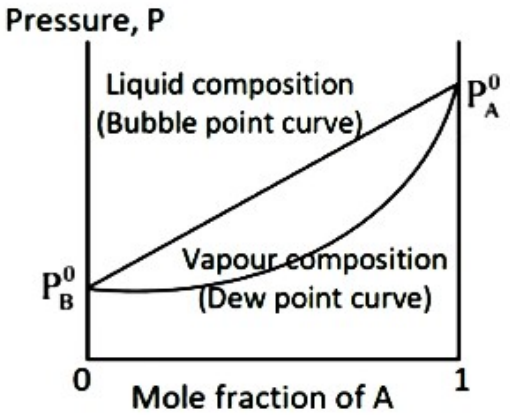
$$= \frac{y_A}{P_A^0} + \frac{1 - y_A}{P_B^0}$$

$$P_T = \frac{P_A^0 P_B^0}{P_A^0 + (P_B^0 - P_A^0) y_A}$$

From the above equation, the plot of P_T with respect to y is a curve instead of a straight line. We can superimpose this curve on the diagram for graphical representation of Raoult's Law to get following phase diagrams.

Pressure versus Composition Phase Diagram (At constant T): P vs. x and y

In this kind of phase diagram, the temperature has a fixed value. The mole fraction of one component is plotted on the horizontal axis and the pressure is plotted on the vertical axis. If A is more volatile than B, then we get following diagram.



Let's understand

Explanation :

(i) On horizontal-axis we have plotted both x and y that is both liquid composition and vapour composition.

(ii) The lower curve (Dew Point Curve) represents the total pressure as a function of the composition i.e. mole fraction in the vapor phase at equilibrium with the liquid phase. It is plotted using P_T vs. y_A equation:

$$P_T = \frac{P_A^C P_B^0}{P_A^0 + (P_B^C - P_A^C)y_A}$$

(iii) The upper curve i.e. Bubble Point Curve (a straight line in the case of an ideal solution) represents the total pressure as a function of composition i.e. mole fraction in the liquid. It is plotted using P_T vs. x_A equation:

$$P_T = (P_A^C - P_B^0)\, x_A + P_B^0$$

(iv) The area between these two curves is vapour-liquid equilibrium region. Vapours cannot exist above the bubble point curve and liquid can not exist below the dew point curve.

(v) Suppose, initially the pressure over the solution is very high so that no vapour exist above the liquid. As we gradually decrease the pressure, a point (Bubble point) comes when we cross the Bubble-Point curve and first bubble of vapour starts forming. Hence, the name bubble point curve. Now we have entered the vapour-liquid equilibrium region. On further decreasing the pressure, a point (Dew Point) comes when we cross the Dew-Point curve when almost all the liquid has evaporated into vapour i.e. only the last drop of liquid (Dew) remains. Beyond this point no liquid exists in the system.

Q14 **Benzene and toluene form nearly ideal solutions. At 300 K, $P°_{toluene}$ = 30 mmHg and $P°_{benzene}$ = 100 mmHg.**

3 mol of toluene and 2 mol of benzene are mixed to make a liquid mixture.

(a) If the pressure over the mixture at 300 K is reduced. At what pressure does the first vapour form?

(b) What is the composition of the first trace of vapour formed ?

(c) If the pressure is reduced further, at what pressure does the last trace of liquid disappear?

(d) Find the composition of the last trace of liquid.

Sol. (a) $P = X_A P^0_A + X_B P^0_B = \frac{3}{5} \times 30 + \frac{2}{5} \times 100 = 58$ mm Hg

(b) Composition of the first trace of vapour formed

$$Y_{benzene} = \frac{X_b P^0_b}{P_{Total}} = \frac{2}{5} \times \frac{100}{58} = \frac{20}{29} \text{ \&}$$

$$Y_{toluene} = 1 - \frac{20}{29} = \frac{9}{29}$$

(c) The last trace of liquid disappear at $Y_A = 3/5$ & $Y_B = 2/5$

$$\frac{1}{p} = \frac{Y_A}{P^0_A} + \frac{Y_B}{P^0_B}$$

$$\frac{1}{p} = \frac{3}{5 \times 30} + \frac{2}{5 \times 100} \quad \text{or}$$

$$p = \frac{250}{6} \text{ mm Hg}$$

(d) The composition of the last trace of liquid

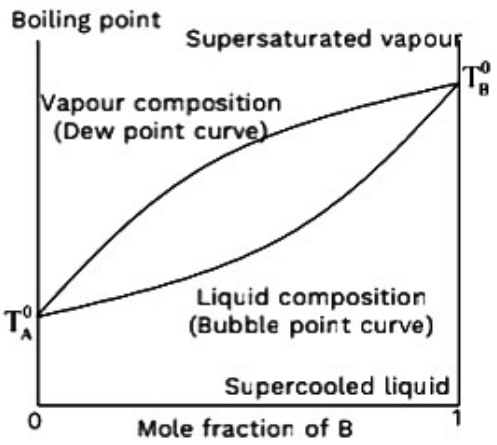
$$X'_{benzene} = \frac{Y_{benzene} P_{Total}}{P^*_b} = \frac{\frac{2}{5} \times \frac{250}{6}}{100} = \frac{1}{6}$$

so $X'_{toluene} = \frac{5}{6}$.

Temperature versus Composition Phase Diagrams i.e. T vs x and y diagram

(i) In this type of phase diagram the pressure is held fixed. The mole fraction of one component is plotted on the horizontal axis and the boiling temperature is plotted on the vertical axis.

(ii) The upper curve (Dew-Point curve) gives the boiling temperature at the given pressure as a function of the mole fraction in the vapour phase, and the lower curve (Bubble-Point curve) gives the boiling temperature at the given pressure as a function of the mole fraction in the liquid phase.



Note: If a liquid has a high vapour pressure at a particular temperature, it means that its molecules can escape easily from the surface. The liquid with the higher vapour pressure at a particular temperature is the one with the lower boiling point. If A is more volatile than B, then we get following diagram.

Distillation:

(i) The compositions of the liquid and vapour that are in mutual equilibrium are not

necessarily the same. Common sense suggests that the vapour should be richer in the more volatile component.

(ii) In a simple distillation, the vapour over a boiling mixture is withdrawn and condensed in a separate container. The liquid collected in the new container is called condensate or distillate and the liquid remaining in the original container is called residue. The condensate has higher mole fraction of the more volatile component than the original mixture. The residue has higher mole fraction of the less volatile component than the original mixture. This is the principle of distillation and is used to separate a more volatile liquid from a less volatile liquid.

(iii) In fractional distillation, the boiling and condensation cycle is repeated successively to get a distillate which gets richer in the more volatile component after each cycle.

2. Completely Immiscible Liquids and Steam Distillation

For mixture of two completely immiscible liquids, each liquid exerts its own vapor pressure, independent of the other, and the total vapor pressure is the sum of the separate vapour pressures of the two components in the pure state at the given temperature.

Immiscible Liquids :

When two liquids are mixed in such a way that they do not mix at all then both liquids constitute two different phases. Hence, **mole fraction of each liquid is one in it's own phase.**

$$P_T = P_A + P_B$$

$P_A = P_A^0 x_A$ But, $x_A = 1$. Thus, $P_A = P_A^0$

$P_B = P_B^0 x_B$ But, $x_B = 1$. Thus, $P_B = P_B^0$

$$P_T = P_A^0 + P_B^0$$

$$\frac{P_A^0}{P_B^0} = \frac{n_A}{n_B}$$

$$\frac{P_A^0}{P_B^0} = \frac{W_A M_B}{M_A W_B}$$

$$P_A^0 = \frac{n_A RT}{V} \; ; \; P_B^0 = \frac{n_B RT}{V}$$

where n_A and n_B are the numbers of moles of each component present in the vapor phase.

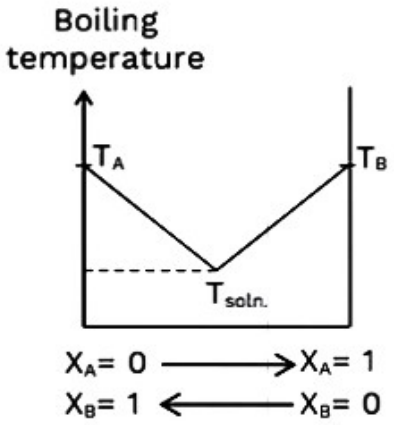
(i) To express the ratio of A to B in the vapor in terms of the actual weights w_A and w_B, the numbers of moles must be multiplied by the respective molecular weights M_A and M_B ; hence,

Steam Distillation of Immiscible liquids : The boiling point of a solution of two immiscible liquids is less than the individual boiling points of both the liquids.

This concept is used in steam distillation.

(i) A system of two immiscible liquids will boil when the total vapor pressure P is equal to the atmospheric pressure. The boiling point of the mixture is thus lower than that of either constituent.

(ii) Further, since the total vapour pressure is independent of the relative amounts of the two liquids, the boiling point, and hence the composition of the vapour and distillate, will remain constant as long as the two layers are present.



A liquid (generally organic) that is immiscible with water, and that has a higher boiling point than water can be boiled (distilled) at a much lower temperature by passing steam through it. In this way, the organic liquid can be purified from impurities using steam distillation.

Q15 **Boiling point of a mixture of water and nitrobenzene is 99°C, the vapour pressure of water is 733 mm of Hg and the atmospheric pressure is 760 mm of Hg. The molecular weight of nitrobenzene is 123. Find the ratio of weights of the components of the distillate.**

Sol. $\frac{W_W}{W_N} = 3.973$

$$\frac{\Delta P}{P_s} = \frac{n_N}{n_w} = \frac{w_N}{123} \times \frac{18}{w_w}$$

$$\frac{27}{733} = \frac{18}{123} \times \frac{w_N}{w_w}$$

$$\Rightarrow \frac{w_w}{w_n} = \frac{18 \times 733}{27 \times 123} = 3.973$$

Azeotropic Mixtures

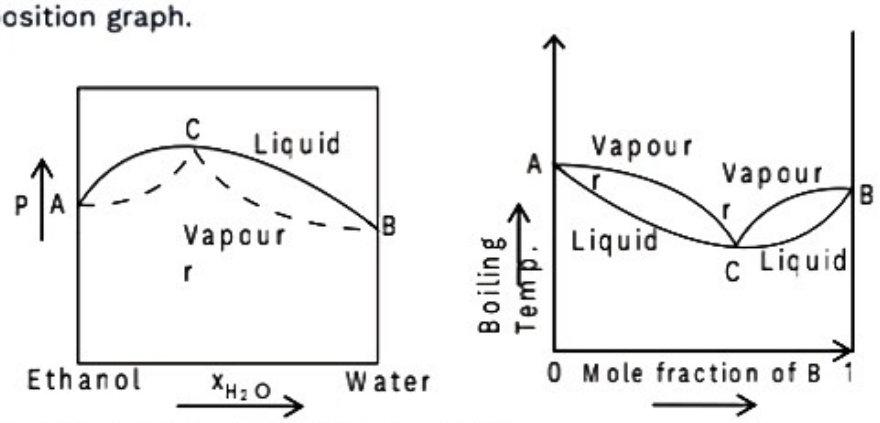
Definitions

Binary mixtures that have same composition in liquid and vapour phase and boil at a constant temperature and can be distilled unchanged in composition are known as azeotropic mixture or simply azeotropes. Thus azeotropes behave as pure liquid. There are two types of azeotropes called as minimum boiling azeotropes and maximum boiling azeotropes respectively.

Types of Azeotropic mixtures

(i) Minimum boiling Azeotropic mixtures

The non-ideal binary solution which shows positive deviation from Raoult's law form minimum boiling azeotropes at a constant composition. As we can see from vapour-pressure-composition graph.



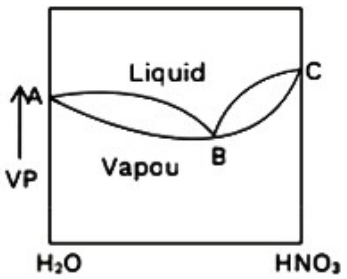
Vapour pressure-composition graph

Minimum boiling point Azeotropic

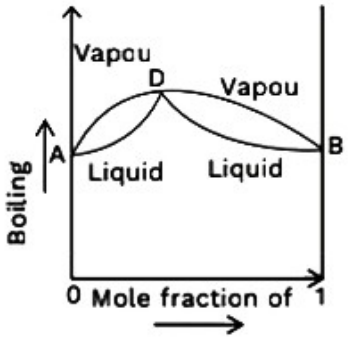
Mixture	% Composition of azeotrope	Boiling point (pressure = 1 atm)
1. Water-Ethanol	96 Ethanol	78.15°C
2. Pyridine-Water	57.00 Pyridine	92.60°C
3. Ethanol-Benzene	32.4 Ethanol	67.80°C
4. Acetic acid-Toluene	28.0 Acetic acid	105.40°C

(ii) Maximum boiling azeotropic mixtures

The non-ideal solution which shows negative deviation Raoult's law form maximum boiling azeotropes. From the vapour-pressure-composition diagram, we can see that the solution has a minimum vapour pressure at a particular composition.



Vapour pressure-composition diagram



Maximum boiling point Azeotropic

Mixture	% Composition of azeotrope	Boiling point (pressure = 1 atm)
1. Nitric acid-Water	68% Nitric acid	125.5°C
2. Acetic acid-Pyridine	65% Pyridine	139.0°C
3. Chloroform-Acetone	80% Chloroform	65.0°C
4. Hydrogen chloride-Water	79.8 Water	108.6°C

Colligative Properties:

Definitions

Those properties of a dilute solution which depends upon the relative number of solute particles in a given volume of the solution or the mole fraction of the solute are called colligative properties.

colligative: from Latin: co means together, ligare means to bind.

The following four properties are colligative properties of solution :
(i) Relative lowering of vapour pressure.
(ii) Elevation in boiling point.
(iii) Depression in freezing point.
(iv) Osmotic pressure.

Colligative property ∝ No. of particles/molecules/ions

or

Colligative property ∝ No. of moles of solute

or

Colligative property ∝ Mole fraction of solute

or

Colligative property ∝ 1/mol. wt. of solute

1. **Relative lowering in vapour pressure :**

Definitions

When a non-volatile solute 'A' is dissolved in a pure solvent 'B', the vapour pressure of the solvent is lowered i.e. the vapour pressure of a solution is always lower than that of pure solvent, because the escaping tendency of solvent molecules decreases.

Do You Know?

Sea water evaporates a little slower than drinking water because sea water contains solute particles which lower the vapour pressure and make evaporation slow.

The lowering of vapour pressure can be understood well by using Raoult's law assuming solute particle is non-volatile and has zero vapour pressure.
If at a certain temperature P° is the vapour pressure of pure solvent, and P_s is the vapour pressure of solution then

Lowering of vapour pressure = $P^\circ - P_s$

Relative lowering of vapour pressure = $\frac{P^0 - P_S}{P^0}$

Now, using Raoult's law

$$P_s = P^0(1-X_A)$$

Where X_A is mole fraction of non-volatile solute

$$\Delta P = P^\circ - P^\circ (1-X_A) = P^\circ(X_A)$$

$$\boxed{\frac{P^0 - P_S}{P^0} = \frac{\Delta P}{P^0} = \frac{n_A}{n_A + n_B} = X_A}$$

also $\boxed{\frac{\Delta P}{P_S} = \frac{n_A}{n_B} = \frac{w_A}{m_A} \times \frac{m_B}{w_B}}$

Points to remember

For a more dilute solution $n_A << n_B$, So
$n_A + n_B \approx n_B$

[Only for very dilution solutions]

Q16 **Calculate weight of urea which must be dissolved in 400 gm of water so final solutions has V.P 2% less than V.P. of pure water:**

Sol. Let V.P be V. of water

$P^\circ - P_s = .02\ V$

$P_s = 0.98\ V$

$\frac{0.02}{0.98} = \frac{w}{60} \times \frac{18}{400}$; where w = weight of urea.

$w = \frac{2 \times 60 \times 400}{18 \times 98}$ gm. = 27.2 g

Q17 **In 80 gm of acetone, 10 gm of a solute is dissolved. V.P. of this sol = 271 mm of Hg.**
If V.P. of pure acctone is 283 mm of Hg. Calculate molar mass of solute.

Sol. $\frac{P^0-P_s}{P_s} = \frac{w}{m} \times \frac{M}{W}$

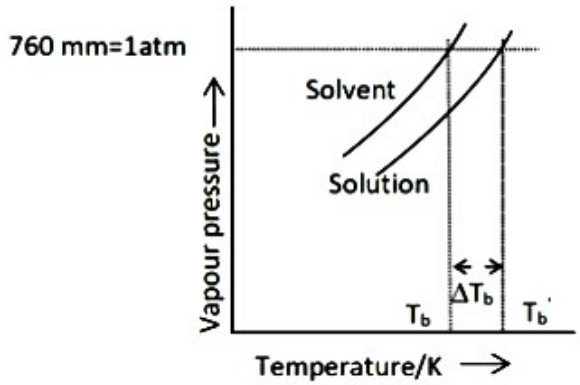
$\frac{283-271}{271} = \frac{10}{m} \times \frac{58}{80}$

$m = 163.7$ gm/mol.

2. Elevation in boiling point (Ebullioscopy):

Introduction

The boiling point of a liquid is defined as the temperature at which its vapour pressure becomes equal to the atmospheric pressure. i.e. 760 mm of Hg.



Do You Know?

Adding salt while boiling your food may make the process efficient, and that is due to the fact that adding any nonvolatile solute like salt elevates boiling point of water and food cooks well before boiling starts. This can be understood well by colligative property Elevation in boiling point

The vapour pressure curve for pure solvent lies above the vapour pressure of solution.

ΔT denotes the elevation of boiling point of a solution.

K_b may be defined as the elevation in boiling point produced when 1 mole of solute is dissolved in 1000 g of the solvent.

$$\therefore \boxed{\Delta T_b = \text{molality} \times K_b}$$

The elevation in boiling point of solution of non-electrolyte is proportional to its molality and equimolal solution of all the substances in the same solvent will show equal elevation in boiling points. These are known as Raoult's laws of elevation of boiling point.

Points to remember

Molal elevation constant is characteristic of a particular solvent and can be calculated from the thermodynamical relationship.

$$K_b = \frac{RT_b^2}{1000L_v}$$

where, R is molar gas constant = 2 cal/mol-K
T_b is the boiling point of the solvent (in K) and L_v the latent heat of vaporisation of pure solvent in calories per gram =

For water $K_b = \frac{2 \times (373)^2}{1000 \times 540} = 0.515$ K-kg/mol

The molal elevation constant for some common solvents are given in the following table

Solvent	B.P. (°C)	Modal elevation constant
Water	100.0	0.52
Acetone	56.0	1.70
Chloroform	61.2	3.63
Carbon tetra chloride	76.8	5.03
Benzene	80.0	2.53
Ethyl alcohol	78.4	1.20

Q18 **1.22 g of a monobasic acid is dissolved in 100 g of benzene. Boiling point of solution increases by 0.13°C with respect to pure benzene. Find the molecular mass of acid in benzene solvent (in u). (K_b of benzene = 2.6 K kg mol^{-1}).**

Sol. 244

$\Delta T_b = K_b \times m$

$0.13 = 2.6 \times \frac{1.22 / M}{0.1}$

$\therefore M = 244$ u

Depression in freezing point (Cryoscopy):



Do You Know?

One of the common ways to clean ice in areas where snow accumulates on streets is spreading salt. This is one of the applications of colligative property, Depression in Freezing point. Adding nonvolatile solute depresses freezing point of a liquid and it freezes below the expected temperature or it melts out if already frozen.

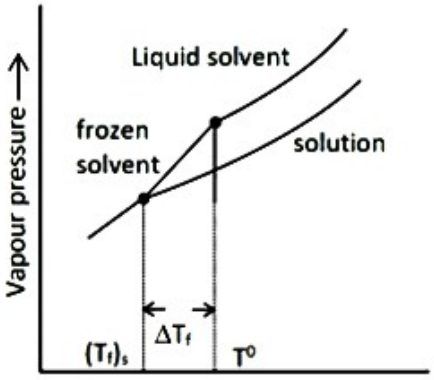
Introduction:

The temperature at which the liquid and its solid state exist in equilibrium with each other is called the freezing point of that liquid.

Or

Temperature at which the solid and liquid states of a given substance have the same vapour pressure.

Temperature/K→



K_f is characteristic constant of a particular solvent and can be calculated from the following relationship.

$$\text{or} \quad \Delta T_f = \frac{1000 K_f \times w_A}{m_A \times w_B}$$

$$\text{or} \quad \boxed{\Delta T_f = \text{molality} \times K_f}$$

K_f is characteristic of a particular solvent and can be calculated from the thermodynamical relationship

$$\boxed{K_f = \frac{RT_f^2}{1000L} = \frac{0.002T_f^2}{L_f}}$$

Where, T_f is the freezing point of solvent in absolute scale and L_f the latent heat of fusion in calories per gram of the solvent. For water,

$$K_f = \frac{0.002 \times (273)^2}{80} = 1.86 \text{ K-kg/mol}$$

the molal depression constant for some common solvents are given in the following table

Solvent	F.P.(°C)	Molal depression solvents
Water	0.0	1.86
Ethyl alcohol	– 114.6	1.99
Chloroform	– 63.5	4.79
Carbon tetra chloride	– 22.8	31.8
Benzene	5.5	5.12
Camphor	179.0	39.70

Q19 **A 100 grams solution of water with x mole of benzene in it, freezes at -0.2°C. The freezing point goes down to 0.25°C as ice separates. How many grams of ice would have separated?**

(A) 18 grams **(B) 20 grams** **(C) 25 grams** **(D) 23 grams**

Sol. (i) $\Delta T_f = m \times K_f$ (initally)

$$0.2 = \frac{X \times 1000}{100} \times 1.86 \qquad X = \frac{0.2}{10 \times 1.86}$$

$\Delta T_f = m \times K_f$ (after freezing till F.P. reaches 25°C)

$\Delta T_f = \frac{X \times 1000}{(100 - y)} \times 1.86$ $\Delta T_f = 0.25$ (Assuming y mass of water separates out)

On solving, Amount of ice y = 20 g ice

Q20 **Calculate the freezing point of a solution of a non-volatile solute in an unknown solvent of molar mass 30 g/mole having mole fraction of solvent equal to 0.8. Given that latent heat of fusion of solid solvent = 2.7 kcal mol^{-1}, freezing point of solvent = 27°C and R = 2 cal mol^{-1} k^{-1}.**

Sol. 10.33°C

$$K_f = \frac{MRT_f^2}{1000\,\Delta H_f}$$

$$= \frac{2 \times 300 \times 300 \times 30}{1000 \times 2700} = 2.00 \text{ K kg mol}^{-1}$$

mole fraction of water = 0.8

mole fraction of solute = 0.2

$$0.2 = \frac{n}{n+N} \text{ and}$$

$$0.8 = \frac{N}{n+N}$$

$$\frac{n}{N} = \frac{0.2}{0.8} = \frac{1}{4} = \frac{wM}{mW}$$

$$\frac{1}{4} = \frac{w \times 30}{mW}$$

$$\therefore \quad \frac{w}{mW} = \frac{1}{4 \times 30}$$

$$\Delta T_f = \frac{1000 \times w \times K_f}{m \times W} = \frac{1000 \times 2.0}{4 \times 30} = 16.67$$

$\therefore$ Freezing point of solution. = **10.33°C.**

Osmosis and osmotic pressure :

Do You Know?

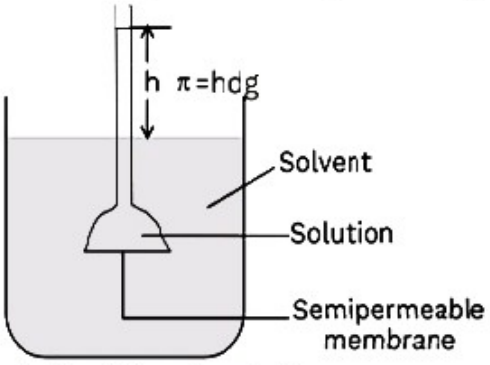
Grapes shrinks in sugar solution because when the grape is placed in sugar water, a hypertonic solution, water will move down the concentration gradient from high concentration (in the grape) to low concentration (the solution), causing it to shrink.

Definitions

Osmosis is defined as the spontaneous **flow of solvent** molecules through **semipermeable membrane** from a high concentration to low concentration range.

Osmosis:

There are many phenomena observe in nature or at home which may be defined as osmosis. For example, making pickle with raw mangoes, they shrivel when put in brine (salt water); when we put wilted flowers in fresh water they revive; when suspended in saline water, blood cells collapse and many more. All these substances are bound by membranes, this is one thing common in all these phenomena. These membranes can be of vegetable or animal origin, they may also be synthetic such as cellophane. They appear to be continuous sheets or films, yet they contain a network of sub microscopic holes. Bigger solute molecules can not pass through these membrane, but small solvent molecules can pass through these holes. Such membranes are called semipermeable membranes (SPM). From pure solvent to the solution, the solvent molecules will flow through the membrane, if this membrane is placed between the solvent and solution. This process is called osmosis.



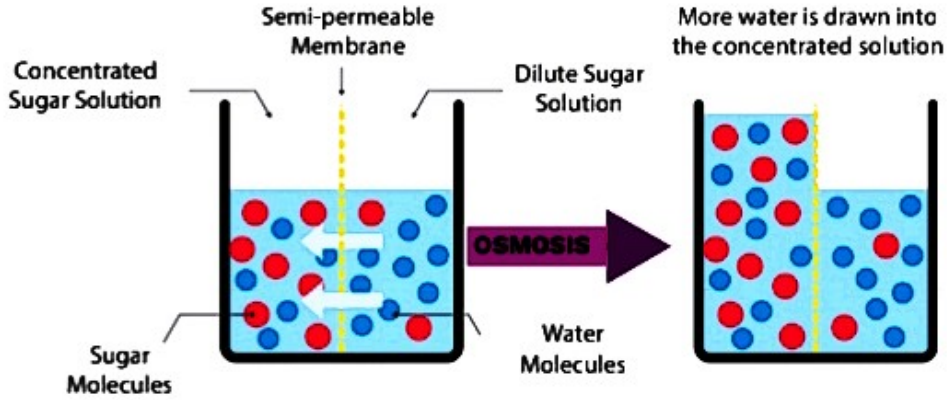
Level of solution rises in the funnel due to osmosis of solvent

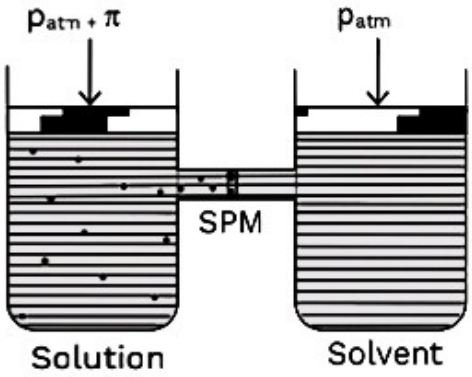
Osmotic pressure (π) **:**

The measure of osmotic pressure of the solution is the hydrostatic pressure developed as a result of osmosis.

or

The hydrostatic pressure built up on the solution which just stops the osmosis. osmotic pressure = hydrostatic pressure.





π = hdg

h = increase in level in the tube of unit cross section.

d = density of solution,

g = acceleration due to gravity

Van't Hoff law for dilute solutions:

The osmotic pressure (P or π) of a solution is directly proportional to its concentration(C) when the temperature is kept constant. The concentration of the solution containg in one gram mole in V litres is equal to 1/V.

Points to remember

Isotonic or iso-osmotic solution: Solutions with the same osmotic pressure are known as isotonic or iso-osmotic $\pi_1 = \pi_2$

i.e. $C_1ST = C_2ST$

$C_2 = C_2$

thus $P \propto C$ (when temperature is constant)

or $P \propto \frac{1}{V}$

or $\boxed{\pi V = \text{constant} = nST}$

Hypertonic solution

A solution having higher osmotic pressure than some other solution is said to be called hypertonic solution.

Semipermeable Membrane

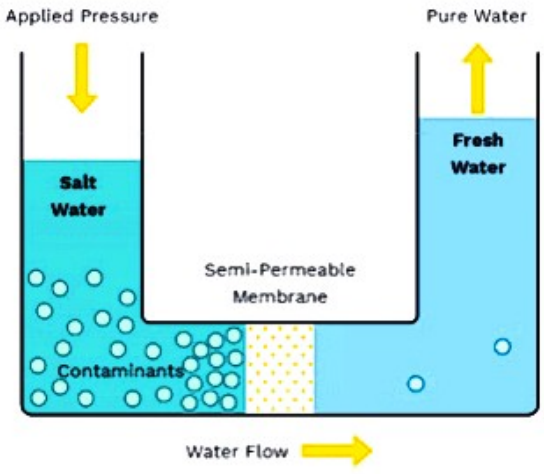
A membrane which allows the passage of solvent molecules but not that of solute.
eg. Copper ferrocyanide $Cu_2[Fe(CN)_6]$.

Reverse Osmosis:

When external applied pressure is greater than osmotic, the flow of solvent molecules can be made to proceed from solution towards pure solvent, i.e., in reverse direction of the ordinary osmosis.
For the desalination of sea water for getting fresh drinking water reverse osmosis is used.

Do You Know?

Reverse osmosis occurs when a pressure larger than
the osmotic pressure is applied to the solution.



Reverse Osmosis



Q21 **Calculate osmotic pressure of 0.1 M urea aqueous solution at 300 K ,**

Sol. $R = 0.082$ lit atm K^{-1}

$\pi = CRT$

$\pi = 0.082 \times 0.1 \times 300$

$\pi = 2.46$ atm.

Q22 **If 10 gm of an unknown substance (non-electrolytic) is dissolved to make 500 mL of solution, then osmotic pressure at 300 K is observed to be 1.23 atm find molecular weight?**

Sol. $1.23 = \frac{10 \times 1000}{M \times 500} \times 0.082 \times 300$

$M = \frac{20}{1.23} \times \frac{0.082}{100} \times 300 = 400$ gm/mol

Q23 **If 6 gm of urea, 18 gm glucose & 34.2 gm sucrose is dissolved to make 500 mL of a solution at 300 K calculate osmotic pressure ?**

Sol. Molecular weight of urea = 60 gm , Glucose = 180gm , Sucrose = 342 gm

$\pi = C \times 0.082 \times 300$

$\pi = \frac{0.3 \times 1000 \times 0.082 \times 300}{500} = 14.76$ atm

Q24 **In a suitable solvent $(C_2H_{10}O_5)_n$ is dissolved and the osmotic pressure p of the solution of various concentration (in kg/m³) is measured at 20°C. The slope of a plot of p against c is found to be 8.134×10^{-3} (SI units) The molecular weight of the insulin (in kg/mol) is :**

(A) 4.8×10^5 **(B) 9×10^5** **(C) 293×10^3** **(D) 8.314×10^5**

Sol. (C) $\pi = CRT$

$\pi = \frac{c}{M}RT$ C = moles/liter, c = kg/m³

$\frac{\pi}{c} = \frac{RT}{M}$ $M = \frac{RT}{\pi / c}$ $[p/c = 8.314 \times 10^{-3}]$

[T = 293 k]

$M = \frac{8.314 \times 293}{8.314 \times 10^{-3}} = 293 \times 10^3$

Q25 For $\underset{(A)}{\text{0.1M urea}}$, $\underset{(B)}{\text{0.1M NaCl}}$, $\underset{(C)}{\text{0.1M } BaCl_2}$ order of P, R.L.V.P, V.P, ΔT_b ΔT_f, ΔT_b, ΔT_f?

Sol.

order of p	C > B > A.
order of R.L.V.P	C > B > A.
order of V.P	A > B > C.
order of ΔT_b	C > B > A.
order of T_b of solution	C > B > A.
order of ΔT_f	C > B > A.
order of T_f of solution	A > B > C.

Q26 **At 10°C, the osmotic pressure of urea solution is 500 mm. The solution is diluted and the temperature is raised to 25°C, when the osmotic pressure is found to be 105.3 mm. Determine extent of dilution.**

Sol. $P \propto CT$

$$\frac{P_1}{C_1T_1} = \frac{P_2}{C_2T_2} \qquad P_1 = 500 \text{ mm} \quad P_2 = 105.3$$
$$C_1 = C_1 \qquad C_2 = ?$$

$T_1 = 10 + 273 = 283$ K

$$\frac{500}{C_1 \times 283} = \frac{105.3}{C_2 \times 298} \qquad C_1 = 1/V_1$$

$$\frac{V_1 \times 500}{283} = \frac{V_2 \times 105.3}{298} \qquad C_2 = 1/V_2$$

$$\boxed{V_2 = 5V_1}$$

Q27 **Dry air was drawn through bulbs containing a solution of 40 grams of urea in 300 grams of water, then through bulbs containing pure water at the same temperature and finally through a tube in which pumice moistened with strong H_2SO_4 was kept. The water bulbs lost 0.0870 grams and the sulphuric acid tube gained 2.036 grams. Calculate the molecular weight of urea.**

Sol. $P^\circ - P_s \propto 0.087$

$P^\circ \propto 2.306$

$$\frac{P^\circ - P_S}{P^\circ} = X_A$$

$$\frac{0.087}{2.306} = \frac{\frac{40}{m}}{\frac{40}{m} + \frac{300}{18}}$$

$$\frac{40}{m}(0.9623) = \frac{300}{18}(0.03772)$$

$$m = \frac{(40)(0.9623)}{0.62879}$$

$m = 61.2$ gm

Abnormal Colligative Properties :

It has been observed that difference in the observed and calculated molecular masses of solute is due to association or dissociation of solute molecules in solution. It results in a change in the number of particles in solution.

Association of solute particles :

The formation of a bigger molecule by the union of two, three or more solute molecules is called association.

Let 'n' simple molecules combine to give an associated molecule as :

$$\underset{\text{(n single molecules)}}{nX} \longrightarrow \underset{\text{(one bigger molecule) } [n > 1]}{(X)_n}$$

The total number of particles in solution becomes less than the number of molecules initially dissolved in the solution and hence the colligative properties will have lower value. As the molar mass of solute is inversely proportional to the colligative properties, so the molar mass of solute will be greater than theoretical value.

Dissociation of solute molecules :

Molecules of electrolytes undergo ionization or dissociation in ionizing solvents to give more particles in solution. For example, one AB molecule ionizes in solution to give two

particles.

$$AB \longrightarrow A^+ + B^-$$

This dissociation results in an increase in the total number of particles, and therefore the value of colligative properties of such solutions will be higher. As the colligative properties are inversely related to molecular weight, so the molecular weight of ionizable solute will be less than the theoretical value.

Van't Hoff Factor (i) :

In order to express the extent of association or dissociation with certain solutes are expected to undergo in solution.
Van't Hoff in 1886, introduced a factor, called Van't Hoff factor (i). The factor i is defined as

$$i = \frac{\text{observed colligative property}}{\text{Calculated colligative property}} = \frac{\text{Normal molecular weight}}{\text{observed molecular mass}}$$

In case of association of solute particles in solution, the observed molecular weight of solute being more than the normal, the value of factor 'i' is less than unity (i.e. $i < 1$), while for dissociation the value of i is greater than unity (i.e. $i > 1$), because the observed molecular weight has lesser value than normal molecular weight.

Van't Hoff factor and degree of dissociation :

The part of the total number of molecules which dissociates in the solution into simple ions or molecules is called the degree of dissociation.

$$i = \frac{\text{number of particles after dissociation}}{\text{Number of particles initially taken.}}$$

Calculation of 'i' :
Let, solute be A_xB_y

$$A_xB_y \longrightarrow x\,A^{y+} + yB^{x-}$$

Initial mol	1	0	0
after dissociation	$1 - \alpha$	$x\alpha$	$y\alpha$

Total no. of solute particles = $1 - \alpha + x\alpha + y\alpha = 1 - \alpha + \alpha\,(x + y)$
or $i = 1 - \alpha + n\alpha$ [where $x + y = n$ (total ions.)]
for strong electrolytes : $\alpha = 1$ or 100% so i = n (total no. of ions)

$$NaCl \rightarrow Na+ Cl^- \quad (i = 2)$$

$$K_2SO_4 \rightarrow 2K^+ + SO_4^{2-} \quad (i = 3)$$

for complex compound

$$K_4[Fe(CN)_6] \rightarrow 4K^+ + [Fe(CN)_6]^{4-} \Rightarrow n = 5$$

S.no.	Solute type	Example	Ionisation	No. of particles in the solution from 1 mole solute (n)	van't Hoff factor ('i')	Abnormal molecular mass
1.	Non-electrolyte	Urea, sucrose, glucose, fructose	—	1	1	m_{normal}
2.	Binary electrolyte AB type	NaCl, KCl, HCl CH_3COOH NH_4OH, NaOH etc.	$AB \rightleftharpoons A^+ + B^-$ $1-\alpha \quad \alpha \quad \alpha$	2	$1+\alpha$	$\frac{m_{normal}}{1+\alpha}$
3.	Ternary electrolyte AB_2 type or A_2B type	$CaCl_2$, $BaCl_2$ H_2SO_4, $K_2[PtCl_6]$	$AB_2 \rightleftharpoons A^{2+} + 2B^-$ $1-\alpha \quad \alpha \quad 2\alpha$ $A_2B \rightleftharpoons 2A^+ + B^-$ $1-\alpha \quad 2\alpha \quad \alpha$	3 3	$1+2\alpha$ $1+2\alpha$	$\frac{m_{normal}}{1+2\alpha}$ $\frac{m_{normal}}{1+2\alpha}$
4.	Quanternary electrolyte AB_3 or A_3B type	$AlCl_3$, $K_3[Fe(CN)_6]$	$AB_3 \rightleftharpoons A^{3+} + 3B^-$ $1-\alpha \quad \alpha \quad 3\alpha$ $A_3B \rightleftharpoons 3A^+ + B^{3-}$ $1-\alpha \quad 3\alpha \quad \alpha$	4 4	$1+3\alpha$ $1+3\alpha$	$\frac{m_{normal}}{1+3\alpha}$ $\frac{m_{normal}}{1+3\alpha}$
5.	General electrolyte AB_{n-1}	One mole of solute giving 'n' ions in the solution	$AB_{n-1} \rightleftharpoons A^{+(n-1)} + (n-1)B^-$ $1-\alpha \quad \alpha \quad (n-1)\alpha$	n	$1+(n-1)\alpha$	$\frac{m_{normal}}{[1+(n-1)\alpha]}$

Van't Hoff factor and degree of association :

The part of the total number of molecules of the substance which is present as associated molecules is known as the degree of association.

$$i = \frac{\text{Number of particles association}}{\text{Number of particles before association}}$$

Calculation of 'i' :

	$nA \rightarrow (A)_n$	
Initial mole	1	0
after dissociation	$1-\alpha$	α/n

$i = 1 - \alpha + \frac{\alpha}{n}$,

α = degree of association

$i = 1 - \alpha + \alpha/n$

$$m_{Abnormal} = \frac{m_{Normal}}{\left[1 - \left(1 - \frac{1}{n}\right)\alpha\right]}$$

Q28 **What must be the concentration of an aqueous solution of common salt if it is to be isotonic with a solution of this substance which freezes at -0.0186°C ? (Assuming NaCl is fully ionized in the first solution).**

Sol. $\Delta T_f = k_f \times m \times i$

$+ 0.0186 = 1.86 \times 2 \times m$

$m = 0.005$ mole/kg

$m = \frac{w}{m} \times \frac{1000}{w}$ $\quad w = 10$

$0.005 = \frac{w}{58.5} \times \frac{1000}{1}$ NaCl

$w = 0.2925$ g

$C = \frac{w}{m} =$ g/lit.

0.2925 g/lit.

Q29 **In which of the following pairs of solutions will the values of the vant Hoff factor be the same?**

(A) 0.05 M K_4 [Fe(CN)$_6$] and 0.10 M $FeSO_4$

(B) 0.10 M K_4[Fe(CN)$_6$] and 0.05 M $FeSO_4 (NH_4)_2SO_4 . 6H_2O$

(C) 0.20 M NaCl and 0.10 M $BaCl_2$

(D) 0.05 M $FeSO_4 (NH_4)_2SO_4 . 6H_2O$ and 0.02 M $KCl . MgCl_2 . 6H_2O$

Sol. (B,D)

Number of particles from K_4 [Fe(CN)$_6$] = 5

number of particles from $FeSO_4 (NH_4)_2 SO_4 . 6H_2O$ = 5

number of particles from $KCl . MgCl_2 . 8H_2O$ = 5

Q30 **Relative decrease in vapour pressure of an aqueous solution containing 2 moles $[Cu(NH_3)_3Cl]Cl$ in 3 moles H_2O is 0.50. On reaction with $AgNO_3$, this solution will form(assuming no change in degree of ionisation of substance on adding $AgNO_3$)**

(A) 1 mol AgCl **(B) 0.25 mol AgCl**

(C) 0.5 mol AgCl **(D) 0.40 mol AgCl**

Sol. (C)

$$\frac{\Delta P}{P} = \frac{ni}{ni+N}$$

$$0.5 = \frac{2i}{2i+3}$$

$i + 1.5 = 2\ i$

$i = 1.5$

$i = 1 + (y - 1)\ \alpha$

$1.5 = 1 + (2 - 1)\ \alpha$

$\alpha = 0.5$

% dissociation of $[Cu(NH_3)_3Cl]Cl$

$\Rightarrow$ 0.5 mole of Cl^- will be formed.

Hence moles of AgCl=0.5

Q31 **$PtCl_4 \cdot 6H_2O$ can exist as a hydrated complex for this 1 molal aq. solution depression in freezing point is 3.72°. Assume 100% ionisation and $K_f(H_2O)$ = 1.86° mol^{-1} kg, then complex is**

(A) $[Pt(H_2O)_6]Cl_4$ **(B) $[Pt(H_2O)_4Cl_2]Cl_2 \cdot 2H_2O$**

(C) $[Pt(H_2O)_3Cl_3]Cl \cdot 3H_2O$ **(D) $[Pt(H_2O)_2Cl_4] \cdot 4H_2O$**

Sol. $\Delta T_f = i \times m \times k_f$

$$i = \frac{3.72}{1.86 \times 1} = 2$$

So salt is $[Pt\ (H_2O)_3\ Cl_3]Cl.\ 3H_2O$.

Q32 **The freezing point of aqueous solution that contains 3% urea, 7.45% KCl and 9% of glucose is (given K_f of water = 1.86 and assume molarity = molality).**
(A) 290 K (B) 285.5 K (C) 267.42 K (D) 250 K

Sol. $\Delta T_f = i.m.\ K_f$

$\Delta T_f = i_1 m_1 K_f + i_2\ m_2\ K_f + i_3\ m_3\ K_f = (m_1 + 2m_2 + m_3)\ K_f$

$$\Delta T_f = \frac{\frac{3}{60} + \frac{7.45 \times 2}{74.5} + \frac{9}{180}}{100} \times 1000 \times 1.86$$

$\Delta T_f = 3 \times 1.86 \quad = 5.58$

T_f of solution = 273 – 5.58 = 267.42 K Ans.

Q33 **If 0.04 M Na_2SO_4 solutions at 300 K is found to be isotonic with 0.05 M NaCl (100 % dissociation) solutions. Calculate degree of dissociation of sodium sulphate ?**

Sol. $i_1C_1RT = i_2C_2RT$

$i_1C_1 = i_2C_2$

$0.04\ (1 + 2\alpha) = 0.05 \times 2$

$\alpha = 0.75 = 75\%$.

Q34 **A storage battery contains a solution of H_2SO_4 38% by weight. At this concentration, vant Hoff factor is 2.50. At what temperature will the battery content freeze? K_f = 1.86 K Kg mol^{-1}.**

Sol. $T_{solution} = k_f \times m \times i$

$0 - T_{solu} = 1.86 \times \times \times 2.V$

$- T_{sol} = 29.04$

$= - 29.04 + 273$

$T_{sol} = 243.21$ K

Q35 **Phenol associates in benzene to a certain extent to form a dimer. A solution containing 20×10^{-3} kg of phenol in 1.0 kg of benzene has its freezing point depressed by 0.69 K. Calculate the fraction of phenol that has dimerised. (K_f for benzene = 5.12 K kg mol^{-1})**

Sol. Observed mol. mass $= \dfrac{1000 \times k_f \times w}{w \times \Delta T}$

$= \dfrac{1000 \times 5.12 \times 20 \times 10^{-3}}{1 \times 0.69}$

$= 148.4$

Normal mol. mass of phenol = 94

$= \dfrac{\text{Normal mol. mass}}{\text{Observed mol. mass}}$

$= 1 - \left[\dfrac{1}{n} - 1\right]\alpha = 1 + \left[\dfrac{1}{2} - 1\right]\alpha$

$= 1 - \dfrac{\alpha}{2}$

$\alpha = 0.733 \times 100$

$= 73.3\%$

Q36 **1000 g of 1 M sucrose solution in water is cooled to -3.534°C. What weight of ice would be separated out at this temperature?**
K_f (H_2O) = 1.86 K Kg mol^{-1}.

Sol. $\Delta T_f = K_f \times m$ m = 1

$\therefore\ m = \dfrac{w}{m} \times \dfrac{1}{W}$

$\Delta T_f = K_f \times \dfrac{w}{m}$ w =1000 g = w = 1kg

$w = \dfrac{K_f}{\Delta T_f} \times \dfrac{w}{m}$ M = 342g = m = 0.342kg

$= \dfrac{1.86 \times 1}{269.4 \times 0.342}$ $\Delta T_f =$ $\because \Delta T_f$ = Temp. of solvent – Temp. of solution

$= 0.02098$ kg

$= \frac{0.02098}{1000} \times 1000$

= 20.98 gram

w = 342 + 20.98 = 352.98 gram.

$= 0 - [-3.53 + 273]$

$= -269.47$

Q37 **A current of dry air was passed through a series of bulbs containing 1.25 g of a solute A_2B in 50 g of water and then through pure water. The loss in weight of the former series of bulbs was 0.98 g and in the later series 0.01 g. If the molecular weight of A_2B is 80 calculate degree of dissociation of A_2B.**

Sol. $\frac{P^\circ - P_s}{P^\circ} = \frac{\text{loss in mass in solvent bulbs}}{\text{Total loss in mass in both sets of bulbs}}$

$= \frac{0.01}{0.98}$

$\frac{1}{98} = \frac{\frac{1.25}{m}}{\frac{1.25}{m} + 2.77m}$ $\quad \left| A_2B \rightleftharpoons 2A^+ + B^- \right.$

m = 44.45

$i = \frac{\text{Normal molecular mass}}{\text{Abnormal molecular mass}}$

$= \frac{80}{44.45} = 1.80$

$\alpha = \frac{i-1}{n-1}$

$= \frac{1.50-1}{3-2} = 40\%$

Q38 **An aqueous solution containing 288 gm of a non-volatile compound having the stoichiometric composition $C_xH_{2x}O_x$ in 90 gm water boils at 101.24°C at 1.00 atmospheric pressure. What is the molecular formula ?**
$K_b(H_2O) = 0.512$ K mol^{-1} kg
$T_b(H_2O) = 100°C$

Sol. $C_xH_{2x}O_x$

$$1.24 = (0.512) \times \frac{\frac{288}{m}}{90} \times 1000$$

$$m = \frac{(288)(1000)(0.512)}{(90)(1.24)}$$

$m = 1321.2$ gm

$12x + 2x + 16x = 1321.2$

$x = 44$

$C_{44}H_{88}O_{44}$

Q39 **The degree of dissociation of $Ca(NO_3)_2$ in a dilute aqueous solution containing 7 gm of the salt per 100 gm of water at 100°C is 70%. If the vapour pressure of water at 100°C is 760 mm. Calculate the vapour pressure of the solution.**

Sol. $a = 0.7$

$n_T = (1 + 2\alpha)\, n_1 + n_2$

$= 2.4\, n_1 + n_2$

$$n_1 = \frac{7}{164}$$

$$n_2 = \frac{100}{18}$$

$$P_s = (760)\, \frac{\frac{100}{18}}{2.4 \times \frac{7}{164} + \frac{100}{8}}$$

$P_s = 746.24$ mm of Hg

Q40 **A 10% solution of cane sugar has undergone partial inversion according to the reaction:**
Sucrose + Water →Glucose + Fructose. If the boiling point of solution is 100.27°C.
(a) What is the average mass of the dissolved materials?
(b) What fraction of the sugar has inverted?
$K_b(H_2O) = 0.512$ K mol^{-1} kg

Sol. Sucrose + water → Glucose + fructose
100.27 – 100 = (0.512) cm

$$\frac{0.27}{0.512} = m = 0.527$$

100 gm → 10 gm
90 gm water $n_{sucrose} = \frac{10}{342}$

$$0.527 = (1+\alpha)\ \frac{10/342}{90} \times 1000$$

$$1 + \alpha = \frac{0.527 \times 90 \times 342}{10 \times (1000)}$$

$\alpha = 0.6225$
62.25%

$$(a) : M_{avg} = \frac{n_i(342)}{n_i(H\alpha)6}$$

$$\frac{342}{1.6225} = 210.78 \text{ gm}$$

Q41 **Sea water is found to contain 5.85% NaCl and 9.50% $MgCl_2$ be weight of solution. Calculate its normal boiling point assuming 80% ionisation for NaCl and 50% ionisation of $MgCl_2$ [$K_b(H_2O) = 0.51$ kg mole^{-1}K].**

Sol. 100 gm → 5.85 gm NaCl
100 gm → 9.5 gm $MgCl_2$
water = 100 – 5.85 – 9.5

= 84.65 gm

$$\Delta T_b = (0.51)\frac{\frac{5.85}{58.5}(1+0.82)+\frac{9.5}{95}(1+0.5)}{84.651}$$

$$T_b - 100 = (0.51)\left[\frac{0.18+0.15}{84.651}\right](1000)$$

$$= \frac{(0.51)(0.33)(1000)}{84.651}$$

$T_b = 100 + 1.988$
$T_b = 101.988°C$

Q42 **Calculate the boiling point of a solution containing 0.61 g of benzoic acid in 50g of carbon disulphide assuming 84% dimerization of the acid. The boiling point and K_b of CS_2 are 46.2°C and 2.3 K kg mol^{-1}, respectively.**

Sol. $\alpha = 0.54$ $\qquad C_6H_5COOH$

$$\Delta T_b = (2, 3)\left(1-\frac{1}{2}(0.84)\right)\left(\frac{0.61}{122/50}\right)\times 1000$$

$$T_b = 46.2 + (2, 3)(0.58)\left(\frac{0.61}{50\times 122}\right)$$

$T_b = 46.33°C$

Q43 **The vapour pressure of a certain liquid is given by the equation :**

$\log_{10}P = 3.54595 - \frac{313.7}{T} + 1.40655 \log_{10}T$ where P is the vapour pressure I mm and
T = Kelvin temperature. Determine the molar latent heat of vaporization as a function of temperature. Calculate the its value at 80 K.

Sol. $\log_{10} P = 3.54595 - \frac{313.7}{T} + 1.40655 \log_{10} T$

$$P = Ce^{-\frac{\Delta H_V}{RT}}$$

$$\frac{\ell nP}{2.303} = 3.54595 - \frac{313.7}{T} + \frac{1.40655}{2.303}\ell n\ T$$

$$= 3.54595 - \frac{313.7}{T} + \frac{1.40655}{2.303}\ln T$$

$$\frac{\Delta H_V}{RT^2}\frac{d(\ell nP)}{dt} = +\frac{313.7}{T^2}(2.303) + \frac{1.40655}{T}$$

$\Delta H_v = R[313.7(2.303) + 1.40655T]$
at T = 80k
$\Delta H_v = (2)\ [313.7(2.303) + 1.40655(80)]$
$\Delta H_v = 16591$ calorie

Q44 **The cryoscopic constant for acetic acid is 3.6 K kg/mol. A solution of 1 g of a hydrocarbon in 100 g of acetic acid freezes at 16.14°C instead of the usual 16.60°C. The hydrocarbon contain 92.3% carbon. What is the molecular formula?**

Sol. $\Delta T_f = 16.6 - 16.14$

$= 0.46°C$

$$0.46 = (3.6)\left(\frac{\frac{1}{M}}{100}\right) \times 1000$$

$$m = \frac{36}{0.46} = 78.26\ gm = = 78.26\ gm$$

$$C = \frac{92.3}{100} \times 78.26$$

$= 72.23$ gm

$$H = \frac{7.7 \times 78.26}{100} = 6.026\ gm$$

$$\frac{72.23}{12} = 6.02$$

$$\frac{6.026}{1} = 6.02$$

C_6H_6

Q45 **A solution of crab heamocyanin, a pigmented protein extracted from crabs, was prepared by dissolving 0.750 g in 125 cm^3 of an aqueous medium. At 4°C an osmotic pressure rise of 2.6 mm of the solution was observed. The solution has a density of 1.00 g/cm^3. Determine the molecular weight of the protein.**

Sol. $$\frac{\left(\frac{0.75}{m}\right)}{125} \times (8.314 \times 10^5 \times 10^2)(277)$$

$$= 2.6 \times 10^{-2}\,(1)(1000)$$

$$m = \frac{(0.75)(8.314 \times 10^7)(277)}{125 \times 26}$$

$$m = 53.14 \times 10^5 \text{ g/mol}$$

Q46 **A storage battery contains a solution of H_2SO_4 38% by weight. What will be the Van't Hoff factor if the $\Delta T_{f\,(experiment)}$ is 29.08. [Given K_f = 1.86 mol^{-1} Kg]**

Sol. $$29.08 = (1.86)\,\frac{\left(\frac{38}{98}\right)}{62} \times 1000\ xi$$

$$i = \frac{29.08 \times 62 \times 98}{1.86 \times 38 \times 1000}$$

$$i = 2.5$$

Q47 **A decimolar solution of potassium ferrocyanide is 50% dissociated at 300K. Calculate the osmotic pressure of the solution. (R = 8.314 $JK^{-1}mol^{-1}$)**

Sol. $K_4[Fe(N)_6] = 1 + 4 \times \frac{1}{2} = 3$

$$C = 1 + 4 \times \frac{1}{2} = 3$$

$$C = \frac{1}{10} M$$

$$\left(\frac{1}{10}\right)_{\times 10^{-3}} \times (8.314)(300)(3)$$

$$\pi = (8.314)(900) \times 100$$

$$\pi = 7.482 \times 10^5 \text{ N/m}^2$$

Additional Questions:
Topic: Raoult's Law

Q48 **Liquids A and B form ideal solution over the entire range of composition. At temperature T, equimolar binary solution of liquids A and B has vapour pressure 45 torr. Keeping the temperature same, a new solution of A and B having mole fractions X_A and X_B, respectively, has v.p. of 22.5 torr. The value of x_A / x_B in the newly formed solution is (Given that the vapour pressure of pure liquid A is 20 Torr at temperature T)**

Sol. 19

$p_{Total} = p_A^\circ \times \chi_A + p_B^\circ \times \chi_B$

and for equimolar solutions $\chi_A = \chi_B = \frac{1}{2}$ Given, $p_{Total} = 45$ torr for equimolar solution

$p_A^\circ = 20$ torr

So, $45 = p_A^\circ \times \frac{1}{2} + p_B^\circ \times \frac{1}{2} = \frac{1}{2}\left(p_A^\circ + p_B^\circ\right)$

or $p_A^\circ + p_B^\circ = 90$ torr

But we know $p_A^\circ = 20$

$p_B^\circ = 90 - 20 = 70$ torr (From Eq.)

so, now, for the new solution from the same formula Given, $p_{Total} = 22.5$ torr So,

$22.5 = 20\chi_A + 70(1 - \chi_A)$

(As $\chi_A + \chi_B = 1$)

or $22.5 = 70 - 50\chi_A$

So, $\chi_A = \frac{70 - 22.5}{50} = 0.95$

Thus $\chi_B = 1 - 0.95 = 0.05 \left(\chi_A + \chi_B = 1\right)$ as

Hence, the ratio

$\frac{\chi_A}{\chi_B} = \frac{0.95}{0.05} = 19$

Q49 **Liquid M and liquid N form an ideal solution. The vapour pressures of pure liquids M and N are 450 and 700 mmHg, respectively, at the same temperature.**

Then correct statement is

x_M = mole fraction of M in solution;

x_N = mole fraction of N in solution;

y_M = mole fraction of M in vapour phase;

y_N = mole fraction of N in vapour phase

(A) $\frac{x_M}{x_N} > \frac{y_M}{y_N}$ **(B)** $\frac{x_M}{x_N} = \frac{y_M}{y_N}$ **(C)** $\frac{x_M}{x_N} < \frac{y_M}{y_N}$ **(D)** $\left(x_M - y_M\right) < \left(x_N - y_N\right)$

Sol. A

The partial vapour pressure, for a solution of volatile liquids of each component of the solution is directly proportional to its mole fraction present in solution. This is known as Raoult's law

Liquid M and N form an ideal solution. Vapour pressures of pure liquids M and N are 450 and 700 mmHg respectively.

$$\therefore p_N^\circ > p_M^\circ$$

So, by using Raoult's law

$$y_N > x_N$$

and

$$x_M > y_M$$

Multiplying (i) and (ii) we get

$$y_N x_M > y_M x_N$$

$$\therefore \frac{x_M}{x_N} > \frac{y_M}{y_N}$$

Thus, correct relation is (a).

Topic: Elevation in Boiling point & Depression in Freezing point

Q50 **75.2 g of C_6H_5OH (phenol) is dissolved in a solvent has $K_f = 14$. Find the percentage of phenol that dimerizes. Given the depression in freezing point is 7 K.**

Sol. Molar mass of solute $(M_B) = \dfrac{1000 \times K_f \times W_B}{W_A \times \Delta T_f}$

$$\Rightarrow M_B = \frac{1000 \times 14 \times 75.2}{1000 \times 7}$$

$M_B = 150.4$ g per mol

Actual molar mass of phenol = 94 g/mol

$$\text{Now, van't Hoff factor, } i = \frac{\text{Calculated molar mass}}{\text{Observed molar mass}}$$

$$\therefore i = \frac{94}{150.4} = 0.625$$

Dimerisation of phenol can be shown as :

$$2C_6H_5OH \rightleftharpoons (C_6H_5OH)_2$$

Initial	1	0
At equilibrium	$1-\alpha$	$\frac{\alpha}{2}$

Total number of moles at equilibrium,

$$i = 1 - \alpha + \frac{\alpha}{2}$$

$$i = 1 - \frac{\alpha}{2}$$

But $i = 0.625$, thus, $0.625 = 1 - \dfrac{\alpha}{2}$

$$\frac{\alpha}{2} = 1 - 0.625$$

$$\alpha = 0.75$$

Thus, the percentage of phenol that dimerises is 75%.

Q51 **Consider the three solvents of identical molar masses. Match their boiling point with their K_b values**

Solvents	**Boiling point**	**K_b values**
X	**100°C**	**0.92**
Y	**27°C**	**0.63**
Z	**283°C**	**0.53**

Sol. Higher the value of K_b of a solvent suggest that there is larger polarity of solvent molecules, which in turn implies higher boiling point due to dipolc-dipole interaction. Therefore, the correct order of K_b . values of the three given solvents is

Solvents	Boiling point	K_b
X	100°C	0.63
Y	27°C	0.53
Z	283°C	0.92

Q52 **The freezing point of benzene decreases by0.45°C if to 20g of benzene 0.2 g of acetic acid is added. Calculate percentage association if acetic acid associates to form a dimer in benzene (K_f for benzene =5.12 K kg mol^{-1})**

(A) 64.6% **(B) 80.4%** **(C) 74.6%** **(D) 94.6%**

Sol. D

Assume the degree of association of acetic acid (CH_3COOH) in benzene is α, then $2CH_3COOH \rightleftharpoons (CH_3COOH)_2$

Inital moles 1 0 at equilibrium $1-\alpha$ $\frac{\alpha}{2}$

$\therefore$ Total moles $= 1-\alpha+\frac{\alpha}{2} = 1-\frac{\alpha}{2}$ or $i = 1-\frac{\alpha}{2}$

Now, depression in freezing point (ΔT_f) is given as $\Delta T_f = iK_f m$

where, K_f = molal depression constant or cryoscopic constant. m = molality

$$\text{Molality} = \frac{\text{number of moles of solute}}{\text{weight of solvent (in kg)}} = \frac{0.2}{60} \times \frac{1000}{20}$$

Putting the values in eq. (i)

$$\therefore \quad 0.45 = \left[1 - \frac{\alpha}{2}\right](5.12)\left[\frac{0.2}{60} \times \frac{1000}{20}\right]$$

$$1 - \frac{\alpha}{2} = \frac{0.45 \times 60 \times 20}{5.12 \times 0.2 \times 1000}$$

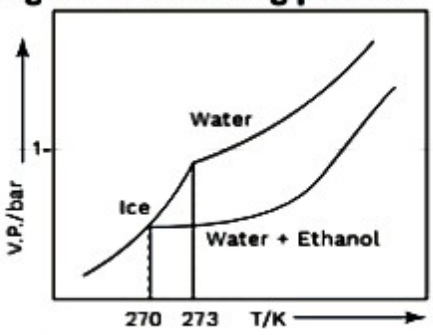
$$\Rightarrow \quad 1 - \frac{\alpha}{2} = 0.527 \Rightarrow \frac{\alpha}{2} = 1 - 0.527$$

$$\therefore \quad \alpha = 0.946$$

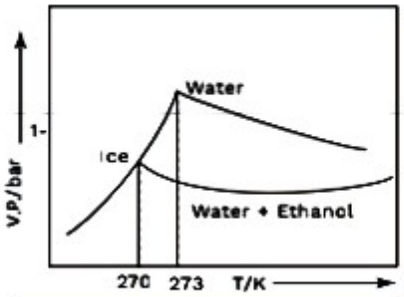
Thus, percentage of association = 94.6%

Q53 **Pure water has freezing point 273 K at 1 bar. To 500 g of water, the addition of 34.5 g of ethanol changes the freezing point of the solution. Use the freezing point depression constant of water as 2 K kg mol^{-1}. The figures shown below represent plots of vapour pressure (V.P.) versus temperature (T). [Molecular weight of ethanol is 46 g mol^{-1}] Among the following, the option representing change in the freezing point is**

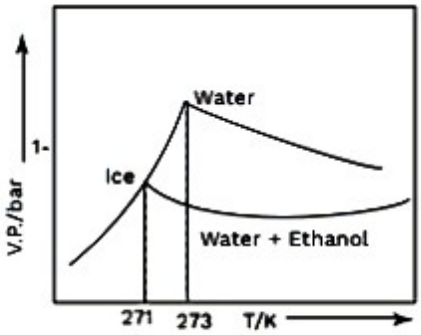
(A)



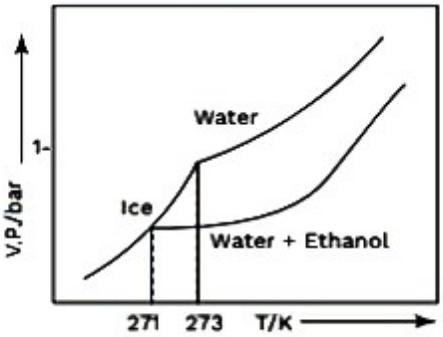
(B)



(C)



(D)



Sol. A

$-\Delta T_f = ik_f m_2$

$= 1 \times 2 \times \frac{34.5}{46 \times 500} \times 1000 = 3$

Vapour pressure curves shown in (a) is in agreement with the calculated value of $-\Delta T_f$. (b) is wrong, vapour pressure decreases on cooling.

Q54 **2.5 g of a non-electrolyte, non-volatile solute in 100 g of water forms a dilute solution, and has elevation in boiling point is 2°C at 1 atm pressure . Assume concentration of solute is much lower than the concentration of solvent, the vapour pressure (mm of Hg) of the solution is (take K_b=0.76 Kkgmol^{-1}).**

(A)740 **(B) 724** **(C) 736** **(D) 718**

Sol. B

The elevation in boiling point is

$\Delta T_b = K_b \cdot m : m = \text{molality} = \frac{n_2}{w_1} \times 1000$

[n_2 = Number of moles of solute, w_1 = Weight of solvent in gram]

$\Rightarrow \quad 2 = 0.76 \times \frac{n_2}{100} \times 1000$

$\Rightarrow \quad n_2 = \frac{5}{19}$

Also, from Raoult's law of lowering of vapour pressure :

$\frac{-\Delta p}{p^\circ} = x_2 = \frac{n_2}{n_1 + n_2} \approx \frac{n_2}{n_1} [\because n_1 >> n_2$

$\Rightarrow -\Delta p = 760 \times \frac{5}{19} \times \frac{18}{100} = 36 \text{mmof Hg}$

$\Rightarrow p = 760 - 36 = 724 \text{mmof Hg}$

Topic: Osmotic Pressure and Osmosis

Q55 **Consider separate solution of 0.500M $C_2H_5OH(aq)$, 0.100MM$g_3(PO_4)_2(aq)$, 0.250MKBr(aq) and 0.125M $Na_3PO_4(aq)$ at 25°C. Which statement is true about these solution, assuming all salts to be strong electrolytes?**
[Assume all electrolytes to be soluble]
(A) They all have the same osmotic pressure
(B) 0.100MM$g_3(PO_4)_2(aq)$ has the highest osmotic pressure
(C) 0.125 M $Na_3PO_4(aq)$ has the highest osmotic pressure
(D) 0.500 M $C_2H_5OH(aq)$ has the highest osmotic pressure

Sol. A
Effective molarity = van't Hoff factor × molarity
0.5M$C_2H_5OH(aq)$ i = 1
Effective molarity = 0.5
0.25MKBr (aq)
i = 2
Effective molarity = 0.5M
0.1 MM$g_3(PO_4)_2$ (aq)
i = 5
Effective molarity
0.125MNa_3PO_4 (aq)
i = 4
Effective molarity = 0.5M
Molarity is same hence, all colligative properties are also same.

Q56 **0.004 M Na_2SO_4 is isotonic with 0.01 M glucose. Degree of dissociation of Na_2SO_4 is**
(A) 75% **(B) 50%** **(C) 25%** **(D) 85%**

Sol. A

For isotonic solutions, they must have same concentrations of ions, Therefore,

$0.004i(Na_2SO_4) = 0.01$

$\Rightarrow i = \frac{0.01}{0.004} = 2.5$

Also $Na_2SO_4 \rightleftharpoons 2Na^+ + SO_4^{2-}$

$1-\alpha \qquad 2\alpha \qquad \alpha$

$\Rightarrow i = 1 + 2\alpha = 2.5$

$\alpha = 0.75 = 75\%$

Q57 **On dissolving 0.6 g of urea (molar mass =60 g mol^{-1} and 1.8 g of glucose molar mass =180 g mol^{-1} in 100 mL of water at 27°C. The osmotic pressure of the solution is (R=0.08206" " L" " atm K^{-1} mol^{-1})**

(A) 8.2 atm **(B) 2.46 atm**

(C) 4.92 atm **(D) 1.64 atm**

Sol. C

For the relation, $\pi = CRT = \frac{n}{V}RT$

Given, mass of urea = 0.6 g Molar mass of urea = 60 g mol^{-1}

Mass of glucose = 1.8 g

Molar mass of glucose = 180 g mol^{-1}

$\pi = \frac{[n_2(\text{urea}) + n_2(\text{glucose})]}{V}RT$

$= \frac{\left(\frac{0.6}{60} + \frac{1.8}{180}\right)}{100} \times 1000 \times 0.0821 \times 300$

$= (0.01 + 0.01) \times 10 \times 0.0821 \times 300$

$\pi = 4.92$ atm

Q58 **13.44 gm of $CuCl_2$ is dissolved in 1 kg of water. Determine the elevation in boiling point of the solution. K_b = 0.5 K Kg mol^{-1}. Molecular wt. of $CuCl_2$ = 134.4**

Sol. Elevation in boiling point of the solution ΔT_b

$\Delta T_b = K_b \times$ Molality

$CuCl_2 \rightarrow Cu^{+2} + 2Cl^-$

1 0 0

1– a α 2a

$i = 1 + 2a$

$i = 3$ $\frac{n}{}$

$= 0.5 \times {}^{W_A} \times i$

$$= 0.5 \times \frac{13.44}{134.1} \times \frac{1000}{1000} \times \frac{1000}{1000} \times 3 = 0.16$$

Rault's Law

Q59 **Vapour pressure of C_6H_6 and C_7H_8 mixture at 50°C is given by P(mm Hg) = 180 X_B = 90, where X_B is the mole fraction of C_6H_6. A solution is prepared by mixing 936g benzene and 736g toluene and if the vapours over this solution are removed and condensed into liquid and again brought to that temperature of 50°C, what would be mole fraction of C_6H_6 in the vapour state?**

Sol. P(mm Hg) = 180 X_B + 90,

P_T = 90 m

$P_B - P_T = 180$

P_B = 270 mm

$$P = 90\left(\frac{736}{92}\right) + 270\left(\frac{936}{2}\right)$$

= 3960

$$y_B = \frac{(270)\left(\frac{936}{78}\right)}{3960}$$

$$y_B = \frac{(3240)}{3960}$$

$$y_t = \frac{90\left(\frac{736}{96}\right)}{3960}$$

$$y_t = \frac{72}{396}$$

$$x_B^{\circ} = \frac{324}{396} \quad x_T^{\circ} = \frac{72}{390}$$

$$P_T^{\circ} = \frac{270 \times 324}{396} + \frac{90 \times 72}{396}$$

$$= \frac{(270)(324) + (90)(72)}{396}$$

$$= 237.27 \text{ mm Hg}$$

$$y_B = \frac{\frac{(270)(324)}{396}}{237.27} = 0.93$$

Q60 **When the mixture of two immiscible liquids (water and nitrobenzene) boils at 372 K and the vapour pressure at this temperature are 97.7 kPa (H_2O) and 3.6 kPa ($C_6H_5NO_2$). Calculate the weight % of nitrobenzene in the vapour.**

Sol. $P_T = P_A^{\circ} + P_B^{\circ}$

$= (97.7 + 3.6)\text{kPa}$

$= (101.3)\text{kPa}$

$$y_A = \frac{P_A^{\circ}}{P_T}$$

$$\left(\frac{w_A}{w_A + w_B}\right) \quad \frac{y_A m_A}{y_A m_A + y_B m_B}$$

$$y_A = \frac{3.6}{101.3}$$

$m_A = 123$

$$y_A = \frac{1 - 3.6}{101.3} \quad m_B = 18$$

$$\text{weight \% age} = \frac{\left(\frac{3.6}{101.3}\right)(123)}{\left(\frac{3.6}{101.3}\right)123 + \left(\frac{97.7}{101.3}\right)18} \times 100$$

$$\frac{(3.6)(123)}{(3.6)(123) + 18(97.7)} \times 100$$

$$= 20.11\%$$

Depression in Freezing point

Q61 **A complex is represented as $CoCl_3.xNH_3$. It's 0.1 molal solution in aq. solution shows $\Delta T_f = 0.558°C$. K_f for H_2O is 1.86 K $mol^{-1}kg$. Assuming 100% ionisation of complex and coordination no. of Co is six, calculate formula of complex.**

Sol. $CoCl_3 \cdot xNH_3$

$$0.558 = (1.86)\,(n)\,(0.1)$$

$$n = \frac{0.558}{0.1(1.86)}$$

$$n = 3$$

$$[Co(NH_3)_5Cl]Cl_2$$

$$x = 5$$

Q62 **1.5 g of a monobasic acid when dissolved in 150g of water lowers the freezing point by 0.165°C. 0.5 g of the same acid when titrated, after dissolution in water, requires 37.5 ml of N/10 alkali. Calculate the degree of dissociation of the acid (K_f for water = 1.86°C mol^{-1}).**

Sol. $$\left(\frac{0.5}{m}\right) = \left(\frac{37.5}{10}\right) \times 10^{-3}$$

$$m = \frac{5}{37.5} \times 1000$$

$$m = 133.33 \text{ gm}$$

$$0.165 = 1.86 \frac{\frac{1.5}{133.3}(1\times\alpha)}{150} \times 1000$$

$$1 + \alpha = \frac{0.165 \times 150 \times 133.33}{1000(1.86)(1.5)}$$

$\alpha = 0.1828$

18.28%

Q63 **If boiling point of an aqueous solution is 100.1°C. What is its freezing point? Given latent heat of fusion and vaporization of water are 80 cal g^{-1} and 540 cal g^{-1} respectively.**
K_f for H_2O = 1.86 K Kg mol^{-1}.

Sol. $\frac{\Delta T_b}{\Delta T_f} = \frac{K_b \times m}{K_f \times m}$

$$\Delta T_f = \frac{K_f}{K_b} \times \Delta T_b \qquad \ldots (1)$$

$k_f = 1.86$ K/m

$$K_b = \frac{R \times T_b^2}{w \times 1000} = \frac{2 \times (100.1+273)^2}{540 \times 1000} = 0.515$$

$$0 - T_s = \frac{1.86}{0.515} \times 100.1 - 100) = \frac{1.86}{0.515} \times 0.1$$

$\boxed{T_s = -0.361°C}$

Relative lowering of vapour pressure

Q64 **A current of dry air is passed through a bulb containing 5 g of a solute in 100 g of water and then through water alone. The losses in weight of the solution and pure water were respectively 0.78 g and 0.02 g. Calculate: (a) relative lowering of vapour pressure and (b) molecular weight of solute.**

Sol.

$$\frac{P^\circ - P_s}{P^\circ} = \frac{\text{Loss in mass of solvent bulbs}}{\text{Loss in mass of solute}}$$

$$= \frac{0.02}{0.78}$$

$$\frac{0.02}{0.78} = \frac{n}{N} = \frac{w}{m} \times \frac{M}{W}$$

$$\frac{0.02}{0.78} = \frac{2.5}{m} \times \frac{18}{1}$$

(b) $\boxed{m = 35.58 \simeq 36}$

$$\frac{P^o - P_s}{P^o} = \frac{2.5}{36} \times 18 = 1.25$$

(a) Relative lowering in vapour pressure = 1.25